ADAPTIVE QUIZ APPLICATION

A Java-Based Personalization Quiz Platform

PROJECT REPORT

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ABSTRACT

Traditional quiz applications commonly present learners with questions selected from a predefined collection, often providing limited adaptation to individual levels of knowledge. Although conventional scoring systems indicate how many questions a learner has answered correctly, scores alone may not adequately communicate the learner's strengths, weaknesses or areas requiring further practice.

The Adaptive Quiz Application, also referred to as AdaptQ, is a Java-based quiz platform designed to support a more personalised learning experience. The application organises questions according to subjects, topics and difficulty levels. Its adaptive mechanism uses learner responses to update estimates of learner ability and question difficulty, helping guide subsequent question selection.

The application uses Java and Spring Boot for backend processing, HTML, CSS and JavaScript for the user interface, and H2 as its default database. Spring Data JPA and Hibernate support database operations, while Apache Maven is used for project building and dependency management.

The system's intended functionality includes adaptive practice, quiz performance tracking, subject-oriented learning and administrative tools for managing and analysing question data. The project also explores additional features such as timed challenges, experience points, achievements and more detailed performance analytics.

This report describes the application's objectives, requirements, technology stack, system design, implementation structure and adaptive question-selection approach. It also discusses testing, observed limitations and possible future enhancements. The report distinguishes the functionality identified in the project from proposed improvements and features whose operation requires further verification.

Keywords: Adaptive Learning, Quiz Application, Java, Spring Boot, Personalised Learning, Question Selection, Performance Analytics.

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CHAPTER 1: INTRODUCTION

1.1 Background

Quizzes are widely used in education to assess learners' understanding of concepts, reinforce previously acquired knowledge and identify areas that require further study. They provide a convenient way to evaluate performance across a variety of subjects and can offer learners immediate feedback on their answers.

Traditional quiz systems frequently rely on predefined question sets and fixed difficulty levels. In such systems, learners with different levels of understanding may receive similar questions, even when their educational needs differ. A beginner may find some questions unnecessarily difficult, while a more experienced learner may find the same questions insufficiently challenging.

Furthermore, an overall score does not always provide a detailed picture of learning progress. Two learners can obtain similar scores while having different strengths and weaknesses across individual topics. Consequently, a quiz application that records only the final score may provide insufficient information to help learners decide which subjects or concepts deserve additional attention.

Adaptive learning systems attempt to address this limitation by adjusting aspects of the learning experience according to information collected about the learner. In an adaptive quiz, factors such as answer correctness, estimated ability and question difficulty can influence the selection of subsequent questions.

The Adaptive Quiz Application, or AdaptQ, explores this approach through a software platform that combines quiz management, performance tracking and adaptive question selection. Its objective is to provide a more personalised practice experience than a quiz system that relies exclusively on fixed question sequences.

1.2 Problem Statement

Conventional quiz applications may provide limited personalisation when selecting questions for learners with different levels of knowledge. Fixed question sequences and generalised difficulty settings can make it difficult to provide an appropriate level of challenge to every user.

Additionally, a single aggregate score may not adequately communicate a learner's performance across different subjects or topics. Without useful performance information, learners may struggle to identify areas that require additional practice.

The problem addressed by this project is the design and examination of a quiz application that uses learner responses to guide question selection and supports the tracking of performance across a structured question bank.

The application seeks to combine adaptive question selection with performance information so that learners can practise relevant material and obtain a clearer view of their quiz results.

1.3 Motivation

The motivation for this project is the opportunity to explore how software can make digital assessment more responsive to individual learners.

A quiz application can do more than display questions and calculate a score. By organising questions according to subject, topic and difficulty, recording learner responses and updating estimates of learner ability, a system can attempt to select questions that better match the learner's current performance.

The project also provides an opportunity to study the integration of several software development concepts. These include backend application development, database persistence, web-based user interfaces, application architecture and the implementation of logic that responds to user input.

By bringing these components together, AdaptQ serves as an example of how a web application can combine conventional quiz functionality with an adaptive mechanism.

1.4 Project Overview

AdaptQ is presented as a Java-based personalised quiz platform. It organises questions by subject, topic and difficulty level and is intended to use learner performance to influence subsequent question selection.

The application uses Java and Spring Boot for backend processing. HTML, CSS and JavaScript provide the web interface, while H2 serves as the default database. Spring Data JPA and Hibernate are used for database operations, and Apache Maven manages dependencies and supports the build process.

The adaptive mechanism is designed around a sequence of related operations. The system selects a question based on estimates of learner ability and question difficulty. It then records the learner's response and relevant performance information. The calibration logic updates its

estimates using the recorded response, and these updated estimates can influence the selection of a subsequent question.

The project also identifies performance tracking and administrative functions as important aspects of the platform. Further capabilities described in the project presentation include timed challenges, experience points, achievements and question-quality monitoring. The implementation status of each feature must be considered separately when evaluating the completed application.

1.5 Project Objectives

The main objectives of the project are as follows:

1. To develop a quiz application that adjusts question difficulty based on learner performance.
2. To provide subject-wise and topic-wise practice across different computing subjects.
3. To record quiz performance and help learners identify areas where additional practice may be beneficial.
4. To explore question selection based on learner responses and estimates of learner ability.
5. To organise quiz questions according to relevant subjects, topics and difficulty levels.
6. To integrate a web interface with backend application logic and database operations.
7. To examine how adaptive calibration can be incorporated into a quiz application.

These objectives describe the intended direction of the project. The extent to which each objective has been achieved will be discussed in the implementation and testing chapters.

1.6 Scope of the Project

The scope of AdaptQ centres on a web-based quiz application with support for question organisation, learner responses and adaptive question selection.

The principal areas covered by the project include:

Question Organisation: Questions are organised using subject, topic and difficulty information, allowing the application to distinguish between different areas of study.

Quiz Interaction: Learners interact with questions through a web interface and submit responses for processing.

Adaptive Question Selection: The application includes logic intended to use estimates of learner ability and question difficulty to guide the selection of questions.

Response Processing: Learner responses are processed so that correctness and other relevant information can inform performance tracking and calibration.

Performance Information: The platform aims to provide information that helps learners understand their quiz performance.

Administrative Functions: The project includes administrative components associated with question data and performance-related information.

The scope of this report includes an examination of the application's architecture, technology stack, implementation structure, adaptive mechanism and observable behaviour.

Features such as AI-assisted question generation, expanded learner progress history and improvements to question management are treated as potential future enhancements unless their implementation can be verified.

1.7 Organisation of the Report

This report is organised into chapters that describe the project from its underlying problem to its implementation and evaluation.

Chapter 1 introduces the project, its motivation, problem statement, objectives and scope.

Chapter 2 discusses the background of quiz applications, adaptive learning and related systems, followed by market and competitor analysis.

Chapter 3 describes the project development process, including the available design decisions, development activities, challenges and limitations.

Chapter 4 presents the functional and non-functional requirements and identifies the intended users and system constraints.

Chapter 5 explains the system design, including the application architecture, workflows, database structure and relevant diagrams.

Chapter 6 describes the technologies used in the application and explains their respective roles.

Chapter 7 examines the application's implementation, including its principal backend components and adaptive question-selection mechanism.

Chapter 8 discusses the user interface and application features, supported by screenshots of the application.

Chapter 9 describes the testing approach, test cases, observed results and limitations of the evaluation.

Chapter 10 presents potential future enhancements, while Chapter 11 concludes the report by summarising the project and its outcomes.

CHAPTER 2: BACKGROUND AND RELATED SYSTEMS

2.1 Introduction

Quiz applications are commonly used to support learning, revision and assessment. They allow learners to answer questions, evaluate their understanding and receive feedback on their performance. The development of web technologies has made it possible to provide these activities through interactive platforms that can store responses and present results to users.

However, the usefulness of a quiz application depends on more than its ability to display questions and calculate scores. The selection of questions, the organisation of learning material and the presentation of performance information can influence how effectively learners use the application.

AdaptQ focuses on adaptive question selection and performance tracking. This chapter introduces the concepts underlying the application, discusses the differences between conventional and adaptive quiz systems, and examines related platforms and their relevance to the project's objectives.

2.2 Traditional Quiz Applications

Traditional quiz applications generally present questions from a predefined collection. Depending on the platform, questions may be selected randomly, arranged in a fixed sequence or organised into quizzes created by an administrator.

When a learner submits an answer, the application evaluates the response and may provide feedback, calculate a score or display a summary of the attempt.

These systems are useful for practice, classroom assessment and revision. They can also provide instructors with a convenient way to distribute questions and evaluate responses.

Despite these advantages, a quiz based on a fixed sequence may not respond to differences in individual learner performance. A learner who consistently answers difficult questions correctly

may continue receiving questions that offer little challenge, while a learner struggling with fundamental concepts may encounter questions that are too difficult.

Furthermore, an overall score may not reveal the specific topics in which a learner requires additional practice. More detailed performance information can therefore be useful when the objective is to support continued learning rather than simply record an assessment result.

These limitations motivate the exploration of quiz systems that use information about learner performance to guide subsequent question selection.

2.3 Adaptive Learning

Adaptive learning refers to an approach in which aspects of an educational experience are adjusted according to information about an individual learner.

In a quiz application, adaptation may involve changing the difficulty of questions, selecting material from particular topics or adjusting the sequence in which questions are presented.

An adaptive quiz can use information collected during an attempt to guide later decisions. For example, a learner who answers several questions correctly may be presented with more challenging material, while repeated incorrect answers may indicate a need for questions at a different difficulty level.

The effectiveness of this approach depends on the quality of the available questions, the accuracy of the information collected and the logic used to interpret learner responses.

Adaptation does not automatically guarantee better learning outcomes. The selection mechanism must be appropriate for the application, and the questions must accurately represent the concepts being assessed.

In AdaptQ, adaptive learning is explored through estimates of learner ability and question difficulty. The system uses these estimates as inputs to its question-selection and calibration processes.

2.4 Personalised Assessment

Personalised assessment aims to make the assessment experience more responsive to differences between learners.

Two learners may receive the same overall score while having different levels of understanding across individual subjects. For example, one learner may perform well in operating systems but struggle with data structures, whereas another learner may show the opposite pattern.

A system that records only an aggregate score may not communicate these differences clearly. Organising questions by subject and topic can provide a more structured way to examine performance.

Personalisation can involve several related activities:

- Selecting questions according to estimated learner ability.
- Organising questions by subject, topic and difficulty.
- Recording answer correctness.
- Examining performance across different areas of study.
- Using previous responses to guide subsequent question selection.

AdaptQ is intended to bring these ideas together within a single application. Its adaptive mechanism focuses on the relationship between learner responses, estimates of ability and question difficulty.

The resulting information can help the system make question-selection decisions and provide a basis for examining learner performance.

2.5 Role of Performance Tracking

Performance tracking is an important component of educational software because it provides information about the outcomes of learning activities.

In a quiz application, useful measurements may include the number of questions attempted, the number answered correctly, overall accuracy and performance across different subjects or topics.

Response time may also provide supplementary information about how learners interact with questions. However, a short or long response time does not independently establish whether a learner understands a concept. It must be interpreted carefully and alongside other information.

Performance tracking can support two related purposes. First, it helps learners review their results and identify areas that may require further practice. Second, it can provide information to the application's adaptive mechanism when that mechanism uses learner responses to update its estimates.

In AdaptQ, response correctness and response-time information are identified as inputs to the application's response analysis and calibration process. The degree to which these measurements improve question selection depends on the implementation and requires evaluation.

2.6 Overview of Related Quiz Platforms

Several established platforms provide digital quizzes, online assessments or interactive learning activities. They offer useful points of comparison for understanding the context in which AdaptQ is positioned.

The following discussion provides a high-level comparison of representative platforms. It is not intended to be an exhaustive feature audit, and the availability of individual features may depend on the platform's subscription plan, configuration or institutional deployment.

2.6.1 Moodle

Moodle is a learning management system used by educational institutions to organise course material, assessments and learning activities.

Its assessment capabilities allow educators to create quizzes, configure questions and review learner results. Its wider learning management functionality distinguishes it from a standalone quiz application.

Moodle is relevant to AdaptQ because both systems address digital assessment and the organisation of learning activities. However, Moodle's principal purpose extends beyond adaptive quiz practice to broader course and learning management.

2.6.2 Quizizz

Quizizz, now operating under the Wayground brand, is associated with interactive quizzes and educational activities. Its platform supports engaging quiz experiences and the review of learner performance.

Its emphasis on interactive participation and educational activities provides a useful comparison for AdaptQ's proposed gamification and quiz features.

AdaptQ differs in its stated emphasis on adaptive question selection based on estimates of learner ability and question difficulty.

2.6.3 Kahoot!

Kahoot! is widely associated with game-based learning, interactive quizzes and participation in live or self-paced activities.

Its game-oriented approach provides a relevant point of comparison for features such as timed challenges, scores and rewards. These concepts can make quiz participation more engaging.

AdaptQ also identifies timed challenges and achievements among its proposed or advertised capabilities. Its distinguishing project objective, however, is the use of an adaptive mechanism to guide question selection.

2.6.4 Comparative Context

The platforms discussed above demonstrate that digital quiz systems can serve different purposes. Some focus on classroom participation and game-based interaction, while others provide assessment tools within a broader learning management environment.

AdaptQ is positioned as a quiz application that explores adaptive question selection and learner performance tracking.

The comparison should not be interpreted as evidence that competing platforms lack adaptive or analytical capabilities. The purpose is to identify the principal emphasis of the project and the types of functionality against which it can be evaluated.

2.7 Market Analysis

Digital educational tools provide learners and educators with ways to organise learning activities, practise concepts and review assessment results.

A quiz application can serve learners who want to revise academic material, prepare for examinations or practise specific technical subjects. Such an application may also be useful in educational environments where structured question banks and performance information are valuable.

The potential usefulness of an adaptive quiz application depends on several factors.

Relevance of Questions: Questions must correspond to the subjects and topics being studied. Poorly written or incorrectly categorised questions can reduce the usefulness of the application.

Quality of Adaptation: Question selection should respond meaningfully to learner performance. A system that changes difficulty without a sound basis may produce an inconsistent learning experience.

Ease of Use: Learners should be able to navigate the application, answer questions and understand their results without unnecessary complexity.

Performance Information: Results should communicate useful information about quiz attempts and, where supported, performance across different topics.

Content Maintenance: Administrators need a practical way to add, review, correct and organise questions as the question bank grows.

Reliability: The application should process responses consistently and preserve relevant data according to its database configuration.

These considerations provide a basis for evaluating the usefulness of AdaptQ and identifying areas for future improvement.

This chapter does not establish market size, revenue forecasts or user demand through quantitative market research. Such conclusions would require supporting evidence, such as published market reports, surveys or documented user studies.

2.8 Competitor Analysis

Competitor analysis helps identify how a project relates to existing applications and clarifies the problem it intends to address.

For AdaptQ, the comparison focuses on the primary purpose of each platform rather than claiming that any competitor lacks a particular feature entirely.*Note: This table presents a high-level conceptual comparison. It does not establish a definitive feature-by-feature audit of current commercial platforms. Exact capabilities should be verified against official documentation before the report is submitted.*

Comparison Criterion	AdaptQ	Moodle	Quizizz / Wayground	Kahoot!
Primary project emphasis	Adaptive quiz practice	Learning management and assessment	Interactive educational quizzes	Game-based learning and quizzes
Question organisation	Subject, topic and difficulty	Course and quiz structures	Quiz and activity structures	Quiz and game structures
Learner performance information	Intended as a core capability	Available through assessment and reporting features	Performance and reporting features	Quiz results and reporting features
Adaptive question selection	A central project objective	Depends on functionality and configuration	Depends on available functionality and configuration	Depends on available functionality and configuration
Gamified interaction	XP, achievements and timed dhallenges are identified in the project	Depends on activities and extensions	A prominent part of the platform's experience	A prominent part of the platform's experience
Intended context	Focused quiz application	Broad educational and course management	Interactive learning and assessment	Interactive and game-based learning

The comparison indicates that AdaptQ's main project focus is adaptive quiz practice. Moodle provides a broader learning management environment, while Quizizz/Wayground and Kahoot! emphasise interactive educational experiences.

AdaptQ's potential value depends on whether its adaptive selection and performance-tracking mechanisms work reliably and provide useful information to learners. The existence of an adaptive mechanism alone does not establish that the application is more effective than established alternatives.

2.9 Opportunities for Improvement

The background and comparison presented in this chapter identify several opportunities relevant to AdaptQ.

First, the quality and organisation of the question bank are important because adaptive selection depends on the availability of suitable questions and accurate metadata.

Second, the adaptive mechanism can be examined to determine whether its estimates respond appropriately to learner performance and whether the selected questions are suitable for the intended difficulty level.

Third, performance information can be improved so that learners can identify patterns in their results rather than relying exclusively on an overall score.

Fourth, administrative workflows can be evaluated to determine whether questions can be added, edited, reviewed and maintained efficiently.

Finally, the application can be evaluated for usability, reliability and consistency. These areas provide practical directions for further development.

2.10 Chapter Summary

This chapter introduced traditional quiz applications, adaptive learning, personalised assessment and the role of performance tracking. It also discussed Moodle, Quizizz/Wayground and Kahoot! as representative platforms in the educational technology landscape.

The discussion established the context in which AdaptQ is positioned: a quiz application that aims to use learner responses and estimates of ability and question difficulty to guide subsequent question selection.

The analysis also highlighted the importance of question quality, reliable adaptation, useful performance information and straightforward administration. These considerations inform the requirements analysis and system design presented in the following chapters.

CHAPTER 3: PROJECT DEVELOPMENT PROCESS

3.1 Introduction

The development of AdaptQ focuses on examining and implementing a quiz application designed to make digital assessment more responsive to individual learner performance. Unlike a conventional quiz application that relies entirely on a fixed sequence of questions, AdaptQ incorporates an adaptive mechanism intended to influence question selection using estimates of learner ability and question difficulty.

The project combines frontend development, backend processing, database management and adaptive question selection. These components work together to support the quiz process, from presenting questions to recording responses and processing information that can influence subsequent question selection.

The project responsibilities are divided into two principal areas. Daniel Fernandes is responsible for backend-oriented work, while Akash Bhatt is responsible for frontend-oriented work. This division provides a clear structure for organising the work associated with the application's different components.

This chapter describes the project's conceptual development, consideration of alternative approaches, selection of the application platform, technology-related decisions, division of responsibilities and technical considerations. It also discusses potential improvements relevant to the application's continued development.

3.2 Project Concept and Initial Planning

The central concept behind AdaptQ is to provide a quiz application in which question selection can respond to learner performance.

A conventional quiz can assess a learner's knowledge by presenting questions and calculating a score. However, a fixed question sequence may not provide an appropriate level of challenge to every learner. Someone who consistently answers questions correctly may benefit from more challenging material, while someone who struggles with particular concepts may benefit from additional practice at a different difficulty level.

An adaptive application attempts to address this problem by using information collected during quiz attempts to guide subsequent question selection.

The concept of AdaptQ combines three principal areas of functionality.

3.2.1 Structured Question Organisation

Questions need to be organised using attributes such as subject, topic and difficulty.

This structure allows the application to distinguish between different areas of study and provides information that can be used when selecting questions for a learner.

For example, a question bank may contain questions from programming, operating systems and data structures. Organising questions by subject and topic makes it easier to identify the material associated with a particular learning area.

Difficulty information is also important because the adaptive mechanism uses estimates of question difficulty when determining which question to present next.

3.2.2 Learner Response Processing

The application must accept learner answers and process the resulting responses.

Response processing involves determining whether an answer is correct and recording relevant information about the attempt. Depending on the implemented functionality, additional information such as response time may also be recorded.

This information supports quiz scoring and provides input to the adaptive mechanism.

3.2.3 Adaptive Question Selection

The adaptive mechanism is a central component of AdaptQ.

The application uses estimates associated with learner ability and question difficulty to guide question selection. After a response is processed, the relevant estimates can be updated, allowing subsequent selection decisions to reflect information collected during the quiz.

The purpose is to make question selection responsive to learner performance rather than relying exclusively on a predetermined sequence.

The effectiveness of this approach depends on the quality of the available questions, the calculations used by the adaptive mechanism and the reliability of the surrounding application.

3.3 Problem Identification and Proposed Solution

The problem addressed by AdaptQ concerns the limitations of quiz applications that do not sufficiently account for differences in learner performance.

A fixed sequence of questions follows a predetermined order regardless of how a learner performs during an attempt. Although this approach is straightforward to implement, it does not inherently respond to changes in the learner's demonstrated understanding.

An overall score can also provide only a limited picture of performance. Two learners may obtain similar scores while having different levels of understanding across individual subjects or topics.

These limitations motivate the development of quiz functionality that combines structured question organisation, response processing and adaptive selection.

AdaptQ approaches this problem by incorporating an adaptive mechanism intended to use learner responses and estimates of ability and question difficulty to guide subsequent questions.

The application also identifies performance tracking as an important supporting capability. By organising quiz material and recording performance information, the system aims to help learners understand their results and identify areas that may require additional practice.

The proposed solution therefore combines conventional quiz functionality with adaptive processing and performance-related information.

Its success depends on whether the adaptive mechanism behaves appropriately, whether the question bank provides suitable material and whether the application processes responses consistently.

3.4 Brainstorming and Consideration of Alternative Approaches

Software design involves selecting an approach that satisfies the project's requirements while remaining feasible within the available resources and development time.

For AdaptQ, several design alternatives are relevant to the choice of question-selection method, application platform, presentation of results and management of question content.

Evaluating these alternatives helps explain the reasoning behind the application's design and identifies the trade-offs associated with different implementation approaches.

3.4.1 Fixed Question Sequences Versus Adaptive Selection

One possible approach is to present questions in a fixed sequence. In this arrangement, the order of questions is predetermined or follows a predefined quiz structure.

A fixed sequence is relatively simple to implement because the application does not need to continually estimate learner ability or adjust question selection in response to answers.

However, this approach provides limited responsiveness to learner performance during an attempt.

An alternative is adaptive selection, in which information about learner responses influences the selection of subsequent questions.

This approach introduces additional implementation complexity because the application must maintain relevant information, update estimates and apply selection rules.

AdaptQ is centred on adaptive question selection. This approach aligns with the project's objective of examining how a learner's performance can influence the difficulty of subsequent questions.

The trade-off is that adaptive selection requires appropriate question metadata, reliable response processing and carefully implemented selection logic.

3.4.2 Aggregate Scores Versus Detailed Performance Information

Another design consideration concerns how quiz results should be presented.

A basic quiz application may display an overall score or the number of correctly answered questions. These measurements are straightforward to understand but may not adequately communicate differences in performance across individual topics.

A more detailed approach can provide additional information about accuracy, subject performance and other relevant measurements.

AdaptQ identifies performance tracking and subject- and topic-oriented practice among its objectives.

Providing more detailed performance information can help learners examine their results beyond a single aggregate score. It can also provide useful information to the application when recorded responses are used by the adaptive mechanism.

However, the usefulness of detailed analytics depends on the accuracy of the underlying records and the clarity of their presentation.

3.4.3 Desktop Application Versus Web Application

During the planning of AdaptQ, the choice between a desktop application and a web application was considered.

Both approaches could support the fundamental requirements of a quiz platform, including question presentation, answer processing, performance tracking and adaptive question selection.

The decision needed to account for the project's requirements and the time available for implementation.

A. Desktop Application Approach

A desktop application would be installed and run on a user's computer.

This approach could provide a self-contained interface for presenting questions, processing answers and displaying quiz results. It could also support a graphical interface for administrative functions.

However, a desktop-based solution would require an appropriate graphical user interface framework, as well as decisions about application packaging and distribution.

Depending on the architecture, additional work might be necessary to support shared data, centralised administration or access from multiple computers.

These considerations made it necessary to evaluate whether a desktop application was the most practical approach for the project's requirements and available development time.

B. Web Application Approach

A web application provides an interface through a browser while processing application logic through a backend.

This approach allows the presentation layer and backend functionality to be organised as separate components. The frontend handles user interaction, while the backend processes requests, performs application operations and interacts with the database.

For AdaptQ, HTML, CSS and JavaScript provide the frontend technologies, while Java and Spring Boot support backend processing.

The web-based architecture provides a straightforward way to organise learner-facing and administrative interfaces without requiring a desktop executable as the primary method of accessing the application.

It also establishes a clear separation between presentation, application logic and data access.

C. Comparison of the Alternatives

The alternatives can be compared using several criteria relevant to the project.

Evaluation criterion	Desktop application	Web application
Access method	Installed application	Browser-based access
Interface technologies	Requires a suitable desktop GUI framework	Can use HTML, CSS and JavaScript
Backend separation	Depends on the chosen architecture	Can separate frontend and backend responsibilities
Distribution	Requires packaging and distribution	Can provide access through a running web server
Administrative interface	Can be included in the desktop application	Can be provided through browser-based pages
Development considerations	GUI design, packaging and application integration	Frontend/backend integration, routing and server configuration
Suitability for this project	Technically possible	Considered more practical within the available time

Both approaches were technically possible. The selection therefore depended on the project's requirements and the practical constraints affecting implementation.

D. Final Decision

The group selected the web application approach because it was considered more practical within the available development time.

The web-based architecture allowed the interface and backend logic to be organised as separate components. It also provided a suitable structure for presenting learner-facing and administrative functionality through browser-accessible pages.

The selected technology stack supported this decision. HTML, CSS and JavaScript provided the frontend, while Java and Spring Boot supported backend processing and application logic.

The decision did not eliminate the need for integration work. Frontend interactions, backend operations, database access and adaptive question selection still needed to work together correctly.

Nevertheless, the web application approach provided a manageable architectural direction for the project.

E. Outcome of the Brainstorming

The desktop-versus-web discussion illustrates how practical constraints can influence software design.

Rather than choosing an architecture solely on the basis of technical possibilities, the project considered the requirements of the application and the time available to develop it.

The web-based approach was selected as a practical way to combine the required interface, backend processing and database functionality. This decision established the architectural direction for the application.

3.4.4 Manual Question Preparation Versus Automated Generation

Question content can be prepared manually or with assistance from automated question-generation tools.

Manual preparation allows questions to be reviewed directly for correctness, clarity and relevance. However, preparing and maintaining a large question bank can require considerable effort.

Automated generation may help administrators prepare draft questions more efficiently, but generated content still requires review. Incorrect answers, ambiguous wording and inappropriate difficulty classifications can reduce the quality of the assessment.

The project presentation identifies AI-assisted question generation as a future enhancement.

Consequently, automated question generation is treated as a possible extension of the application rather than a confirmed capability of the current implementation.

The question bank and its management remain important areas because the adaptive mechanism depends on the availability of suitable questions.

3.5 Selected System Design Approach

The application combines a web-based interface with backend processing and database access.

The frontend provides the user-facing elements through which learners and administrators interact with the application. The backend processes requests, coordinates quiz-related operations and interacts with the persistence layer.

The adaptive mechanism operates within the application's backend logic. It uses information associated with learner responses and question characteristics to guide question selection.

The architecture can be understood through four conceptual layers.

3.5.1 Presentation Layer

The presentation layer provides the user interface through HTML, CSS and JavaScript.

It displays relevant information, accepts user input and provides the interface through which users interact with the application.

The presentation layer is primarily concerned with how information is displayed and how user actions are initiated.

3.5.2 Application Layer

The application layer uses Java and Spring Boot to handle requests and coordinate application operations.

It provides the logic associated with quiz activities, response processing and interactions with other backend components.

This layer connects user actions with the relevant operations performed by the application.

3.5.3 Adaptive Processing Layer

The adaptive processing layer represents the functionality associated with learner ability estimates, question difficulty estimates and question selection.

When a learner submits a response, the adaptive mechanism can use the resulting information to update estimates and influence subsequent selection decisions.

The exact calculations and relationships between the relevant components are examined in the implementation chapter.

3.5.4 Persistence Layer

The persistence layer stores and retrieves application information through the database and associated data-access components.

H2 is identified as the default database, while Spring Data JPA and Hibernate support database operations.

The precise entities, relationships and configuration used by the application are examined in the system design chapter.

These layers provide a conceptual representation of the application's responsibilities. Their exact implementation boundaries should be confirmed against the source code.

3.6 Division of Responsibilities

The project work is divided into backend-oriented and frontend-oriented responsibilities.

This division provides a practical way to organise the work associated with different parts of the application while recognising that both areas must be integrated for the system to function correctly.

3.6.1 Backend Development

Daniel Fernandes is responsible for backend-oriented work.

The backend is responsible for processing requests, executing application logic and interacting with the database. These responsibilities are central to the quiz process because submitted answers and application data must be processed consistently.

The principal backend responsibilities are described below.

Application Logic: Working with the Java components responsible for coordinating quiz-related operations.

Request Processing: Examining how Spring Boot handles requests and invokes the relevant backend functionality.

Database Integration: Working with the data-access components responsible for storing and retrieving application information.

Quiz Processing: Examining the logic associated with question retrieval, answer submission and response processing.

Adaptive Functionality: Understanding the components that estimate learner ability, update question difficulty estimates and influence subsequent question selection.

Backend Evaluation: Examining the behaviour of backend operations and identifying issues that require investigation.

The adaptive mechanism is particularly important to the backend because it relies on calculations and decisions made by the application's logic.

The quality of this functionality depends on the correctness of the implementation, the available question bank and the reliability of the data used during processing.

3.6.2 Frontend Development

Akash Bhatt is responsible for frontend-oriented work.

The frontend provides the interface through which users interact with AdaptQ. It presents information, accepts user input and communicates relevant results in a form that users can understand.

The principal frontend responsibilities are described below.

Interface Structure: Working with the HTML elements that define the organisation of application pages.

Visual Presentation: Working with CSS to control layout, spacing, typography and the appearance of interface elements.

Client-Side Interaction: Working with JavaScript to support interactive behaviour and user actions.

User Navigation: Examining how users move between pages and access quiz-related functionality.

Information Presentation: Ensuring that questions, responses, results and other available information are presented clearly.

Frontend Evaluation: Examining interface behaviour and identifying presentation or interaction issues that require attention.

The frontend contributes directly to the usability of the application. Even when backend processing is correct, confusing navigation or unclear feedback can make the application difficult to use.

3.6.3 Coordination Between Backend and Frontend

The frontend and backend are interdependent components of the application.

The frontend initiates user actions, such as submitting an answer or navigating to a quiz. The backend processes the relevant request and performs the necessary operations. The resulting information is then made available to the interface for presentation.

This interaction requires the two components to exchange information consistently.

For example, submitting an answer requires the application to associate the response with the relevant question and attempt. The backend must process the response and provide the appropriate result, while the frontend must present that result to the learner.

The adaptive mechanism adds another level of processing because the backend may use the submitted response to guide subsequent question selection.

Separating responsibilities helps organise the work, but the complete application must still be evaluated as an integrated system.

3.7 Technology-Related Design Decisions

The technologies used by AdaptQ support different responsibilities within the application.

3.7.1 Java and Spring Boot

Java provides the programming language used for backend processing, while Spring Boot provides the framework for structuring and running the application.

This combination supports the organisation of application logic into components and provides mechanisms for handling web requests and configuring application services.

These capabilities are relevant to quiz management, response processing, database interaction and adaptive question selection.

3.7.2 HTML, CSS and JavaScript

HTML, CSS and JavaScript provide the technologies used for the web interface.

HTML defines the page structure, CSS controls presentation and JavaScript supports interactive behaviour.

Together, these technologies allow the interface to be organised separately from backend processing.

3.7.3 H2, Spring Data JPA and Hibernate

H2 is identified as the application's default database.

Spring Data JPA and Hibernate support data-access operations by connecting Java entities with database records.

These technologies provide a structured approach to storing and retrieving application data.

The database configuration must still be examined to establish whether data is stored in memory or retained between application runs.

3.7.4 Apache Maven

Apache Maven manages the application's dependencies and build configuration.

It supports the process of compiling and packaging the application and provides a consistent way to manage the libraries required by the backend.

The Maven configuration also provides information about the dependencies and runtime requirements of the application.

3.8 Technical Challenges and Design Considerations

An application combining web-based quiz functionality with adaptive processing presents several technical considerations.

3.8.1 Frontend and Backend Integration

The frontend and backend must exchange information in a consistent format.

If the interface submits information that the backend does not expect, an operation may fail or produce an incorrect result. Similarly, backend responses must be interpreted correctly by the frontend.

The application should therefore be examined to determine whether requests, responses and errors are handled consistently.

3.8.2 Database Persistence

Quiz applications depend on the correct storage and retrieval of questions, user information and response records.

The database configuration determines how this information is retained and accessed. Incorrect configuration or persistence behaviour can affect quiz records and performance calculations.

Testing should establish whether relevant information is stored and retrieved as intended.

3.8.3 Adaptive Calibration

The adaptive mechanism updates its estimates using information obtained from learner responses.

An incorrect update may influence subsequent question selection. The calculations therefore require careful examination and testing.

The mechanism should be evaluated using controlled examples in which the expected behaviour can be compared with the application's actual output.

3.8.4 Question Bank Limitations

Adaptive selection depends on the availability of suitable questions.

If the question bank contains too few questions for a particular subject or difficulty level, the application may not always be able to select a suitable next question.

Question quality, topic coverage and difficulty classification are consequently important considerations for the reliability of the system.

3.8.5 Feature Verification

An interface may advertise a feature without the complete functionality being available behind it.

Features such as timed challenges, achievements, performance analytics and administrative tools should therefore be checked against the underlying implementation.

This verification helps distinguish existing capabilities from features that require further development.

3.9 Future Development Considerations

The examination of the application's design identifies several areas that could be developed further.

Improved Question Management: Administrative workflows could be improved to make it easier to add, edit, review and organise questions.

Refined Adaptive Selection: The question-selection mechanism could be evaluated using additional test scenarios and refined where its behaviour does not match the intended design.

Expanded Progress History: The application could provide a more detailed history of learner attempts and performance across topics.

AI-Assisted Question Generation: Automated tools could be used to prepare draft questions for administrator review, provided that generated content is checked for accuracy and suitability.

Improved Evaluation: More extensive testing could examine the consistency of database operations, the reliability of adaptive selection and the usability of the interface.

These improvements are presented as potential future work rather than claims about functionality already implemented.

3.10 Chapter Summary

This chapter described the conceptual development of AdaptQ, the problem it addresses, the consideration of alternative approaches and the selection of a web-based architecture.

The desktop-versus-web comparison established that both approaches were technically possible, but the web application was considered more practical within the available development time. The selected approach supports a separation between the user interface, backend processing and database operations.

The chapter also described the division of responsibilities between Daniel Fernandes, who is responsible for backend-oriented work, and Akash Bhatt, who is responsible for frontend-oriented work.

Finally, the chapter identified technical considerations involving frontend/backend integration, database persistence, adaptive calibration, question-bank quality and feature verification.

These considerations provide a foundation for the requirements analysis presented in Chapter 4, which defines the application's functional and non-functional requirements, intended users and operating constraints.

CHAPTER 4: SYSTEM DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter describes the requirements, architecture, technologies, design, and implementation of AdaptQ, an adaptive quiz application developed using Java and Spring Boot. The application provides subject-based quiz practice and uses information about learner performance to guide the selection of subsequent questions.

The chapter first outlines the system requirements and technology stack, then describes the application architecture, adaptive quiz mechanism, question bank, and principal implementation components. To keep the design representations together and easy to locate, the system diagrams are presented in a dedicated section near the end of the chapter.

4.2 System Requirements

The requirements of AdaptQ focus on providing a usable quiz interface, processing learner responses, and using response information to guide subsequent question selection. The requirements below summarise the principal capabilities and design objectives described for the application. The precise availability of individual features depends on their implementation in the current version.

4.2.1 Functional Requirements

The principal functional requirements of AdaptQ are summarised in Table 4.1.

ID	Requirement	Description

FR-01	User Access	Provide the application's authentication functionality and distinguish relevant user roles.
FR-02	Subject-Based Practice	Organise quiz questions into subjects such as Java, Python, C++, Operating Systems, and Data Structures and Algorithms.
FR-03	Question Presentation	Present questions to learners through the web interface.
FR-04	Response Processing	Process submitted answers and determine whether responses are correct.
FR-05	Adaptive Question Selection	Use learner performance and question-related information to guide the selection of subsequent questions.
FR-06	Performance Tracking	Record response information that can be used to assess learner performance.
FR-07	Question Management	Provide backend support for maintaining the question bank.
FR-08	Administrative Insights	Support administrative views relating to question statistics, difficulty calibration, and learner information.

Table 4.1: Functional Requirements of AdaptQ

These functional requirements describe the main capabilities associated with AdaptQ. They should not be interpreted as confirmation that every listed capability is exposed through a complete user-facing workflow in the current version.

4.2.2 Non-Functional Requirements

The following non-functional requirements inform the design of the application.

Requirement	Description
Usability	The interface should present questions, answer options, and relevant results in an understandable manner.
Maintainability	Separating controllers, services, and data-access components should make the application easier to modify.
Performance	Question retrieval and response processing should be sufficiently efficient for normal quiz interactions.
Data Consistency	Learner responses and question information should be stored and retrieved consistently.
Portability	A web-based interface allows access through a compatible browser without requiring a separate desktop client.
Extensibility	The architecture should allow additional subjects, questions, and features to be introduced as the application develops.

Table 4.2: Non-Functional Requirements of AdaptQ

These are design objectives rather than claims that each requirement has been independently verified through formal performance or usability testing.

4.3 Technology Stack

AdaptQ uses a web application technology stack in which the frontend provides the user interface and a Java-based backend handles application operations. The principal technologies and their purposes are summarised in Table 4.3.

Technology	Purpose
Java	Main programming language used for backend application logic.
Spring Boot	Backend framework used to structure and run the web application.
HTML	Defines the structure of the web interface.
CSS	Controls the visual presentation of the interface.
JavaScript	Supports client-side interactions.
H2 Database	Provides the default relational database configuration.
Spring Data JPA	Provides an abstraction for data-access operations.
Hibernate	Provides object-relational mapping between Java objects and relational database records.
Apache Maven	Manages project dependencies and supports the build process.

Table 4.3: Technology Stack of AdaptQ

The project uses H2 as its default database configuration, which supports local execution without requiring a separate database server. The application can also be configured for MySQL when the appropriate configuration and database environment are provided.

Java and Spring Boot provide the foundation for the backend application, while HTML, CSS, and JavaScript support the web-based interface. Spring Data JPA and Hibernate facilitate database operations and object-relational mapping.

The use of these technologies supports separation between the interface, application logic, and data-access responsibilities.

4.4 System Architecture

AdaptQ follows a web application architecture consisting of a presentation layer, an application layer, and a data-persistence layer. The presentation layer provides the interface through which users interact with the application. The backend receives requests, applies application logic, and coordinates quiz-related operations. The persistence layer stores and retrieves information required by the application.

This separation of responsibilities helps organise the application into components that can be developed and maintained independently.

4.4.1 Presentation Layer

The presentation layer is implemented using HTML, CSS, and JavaScript.

HTML provides the structure of web pages, CSS controls their appearance, and JavaScript supports client-side interactions. Together, these technologies provide the interface through which users navigate the application and interact with quiz functionality.

The frontend communicates with the backend to obtain information and submit user actions. Keeping the interface separate from the main application logic allows the presentation and backend components to be developed and maintained independently.

4.4.2 Application Layer

The backend is implemented in Java using the Spring Boot framework. It handles application requests and coordinates operations such as authentication-related functionality, quiz processing, response handling, question selection, and the processing of performance information.

The backend is organised into controllers and service components. Controllers handle incoming requests, while services implement application operations.

The principal service components described for AdaptQ include:

ResponseService: Processes learner responses and coordinates response-related updates.

CalibrationService: Supports the updating of question-related calibration information.

TestCompositionService: Supports the composition and selection of questions for quiz activities.

This separation helps distinguish request handling from application logic. The exact calls between components should be represented in design diagrams only after checking them against the implementation.

4.4.3 Data-Persistence Layer

AdaptQ uses Spring Data JPA and Hibernate to support database operations.

Spring Data JPA provides an abstraction for data access, while Hibernate serves as the object-relational mapping implementation. These technologies allow application entities to be represented as Java objects and persisted in a relational database.

H2 is the default database configuration. The persistence layer stores information required by the application, including question-bank data and learner-response records.

MySQL can be used as an alternative when the application configuration and database environment are adjusted accordingly.

4.5 Design of the Adaptive Quiz System

The adaptive quiz functionality is a central part of AdaptQ. Unlike a conventional quiz that follows a completely fixed question sequence, AdaptQ uses information about learner performance to guide subsequent question selection.

The implementation includes logic for estimating learner ability and updating question difficulty. Its approach is inspired by rating-based and item-response concepts, using learner responses to inform later selections.

The mechanism should be understood as a simplified adaptive approach rather than a professionally validated computerised adaptive testing (CAT) system.

4.5.1 Learner Performance and Ability Estimation

The system maintains an estimate of learner ability that can be updated as responses are processed.

When a learner submits an answer, the backend evaluates the response and uses the result as an input to the adaptive process. Correctness provides information about how the learner performed on the question.

The system uses this information to update its estimate of learner ability. The resulting estimate helps guide the selection of subsequent questions.

This estimate is intended to support adaptive practice and should not be interpreted as a formally validated measurement of a learner's overall knowledge.

4.5.2 Question Difficulty Calibration

Question difficulty is another factor considered during adaptive selection.

The implementation includes a calibration mechanism that updates question-related estimates based on response information. Response correctness contributes to these updates, while response time is also used in estimating question-related characteristics.

These estimates provide information that can be used to refine subsequent question selection. The quality of the estimates depends on the available response data, the behaviour of the update mechanism, and the quality and coverage of the question bank.

4.5.3 Adaptive Question Selection

The question-composition component uses learner and question information to identify suitable questions.

The implementation considers the estimated learner ability when choosing questions, with a selection target based on the ability estimate and an adjustment factor. It then considers a limited set of questions whose estimated difficulty is close to that target.

The system also attempts to avoid presenting questions that the learner has already encountered, where possible.

At a high level, the adaptive selection cycle consists of the following steps:

1. Obtain the learner's current ability estimate.
2. Identify eligible questions from the available question bank.
3. Calculate a target difficulty using the learner estimate and the selection adjustment.
4. Identify questions with difficulty estimates close to the target.
5. Select a suitable question while attempting to avoid repetition.
6. Present the question to the learner.
7. Process the response and update relevant estimates.
8. Use the updated information to guide later selections.

This cycle allows subsequent question selection to be influenced by the learner's performance rather than relying entirely on a predetermined question order.

4.5.4 Role of Response Time

In addition to correctness, the implementation uses response-time information as part of its question-calibration logic.

Response time can provide supplementary information about how a question behaves during practice. However, a fast response does not necessarily indicate mastery, and a slow response does not necessarily indicate a lack of understanding.

For this reason, response time is treated as one input to the application's internal calculations rather than a definitive measure of learner ability.

The usefulness of this information depends on the quality of recorded response data and the behaviour of the calibration mechanism.

4.6 Question Bank and Subject Organisation

The question bank provides the content required for quiz activities. Questions are associated with subject and difficulty information so that the application can organise practice and support adaptive selection.

The available subject categories include:

- Java
- Python
- C++
- Operating Systems
- Data Structures and Algorithms

This organisation allows the application to support practice across multiple computing subjects rather than restricting users to a single topic.

The backend also includes functionality to populate the question bank when it is empty, supporting the availability of initial question data when the application is run in a suitable environment.

The usefulness of the adaptive mechanism depends substantially on the quality and coverage of the question bank. If a subject contains too few questions, or if difficulty estimates are inaccurate, the range of suitable adaptive selections may be limited.

Question-bank expansion, improved question management, and more systematic calibration are therefore important areas for further development.

4.7 Implementation of the Main Components

The implementation divides the application's responsibilities among controllers, services, and data-access components. This organisation is intended to make individual responsibilities easier to understand and maintain.

4.7.1 Request Handling

The controller layer receives requests from the web interface and directs them to the appropriate backend operations.

This provides a structured way to expose application functionality without placing all processing logic inside the interface. Controllers coordinate with the relevant services to handle quiz-related requests.

4.7.2 Quiz and Response Processing

Quiz-related services coordinate question selection and the processing of learner responses.

When a response is received, the backend determines the outcome and coordinates the relevant updates to learner and question information. The updated information can then be used by the question-selection process.

The separation between response processing and question composition allows these responsibilities to be managed by different service components.

4.7.3 Persistence and Data Access

The data-access layer supports the retrieval and storage of application information.

Spring Data JPA and Hibernate reduce the need to implement repetitive database operations manually. A relational database provides a structured way to represent application entities and preserve information across supported application operations.

H2 is used in the default configuration. Alternative database use depends on the appropriate application configuration and database environment.

4.7.4 Administrative Functionality

The project includes administrative views relating to question-bank information, difficulty calibration, analytics, and learner information.

These capabilities support monitoring and maintenance of the question bank and provide an administrative perspective on the data used by the application.

The exact scope of administrative actions depends on the available interface and backend endpoints. Backend support for an operation should not automatically be interpreted as proof that a complete user-facing workflow exists for it.

4.7.5 Build and Application Configuration

Apache Maven is used to manage dependencies and support the project build process. The Spring Boot configuration defines how the application starts and how its components are configured.

The default H2 database configuration simplifies local execution because a separate database server is not required for the default setup.

MySQL can be used as an alternative when the application configuration and database environment are adjusted accordingly.

4.8 System Design Diagrams

This section brings together the principal diagrams used to represent the architecture and behaviour of AdaptQ.

Each diagram provides a different view of the system. The architecture diagram shows the major layers and components, the activity diagram presents the adaptive quiz workflow, and the sequence diagram shows the messages exchanged during a quiz interaction.

Together, these diagrams provide a visual overview of the system structure and the process followed during adaptive quiz practice. The diagrams should be checked against the implemented application before final submission to ensure that their components, relationships, and message flows accurately reflect the software.

4.8.1 High-Level System Architecture

The high-level system architecture represents the relationship between the web interface, Spring Boot application logic, service components, data-access layer, and database.

The frontend provides the interface through which users interact with the application, while the backend processes requests and coordinates quiz-related operations. The backend service components support question selection, response processing, and question calibration.

Spring Data JPA and Hibernate support communication between the application and the relational database.

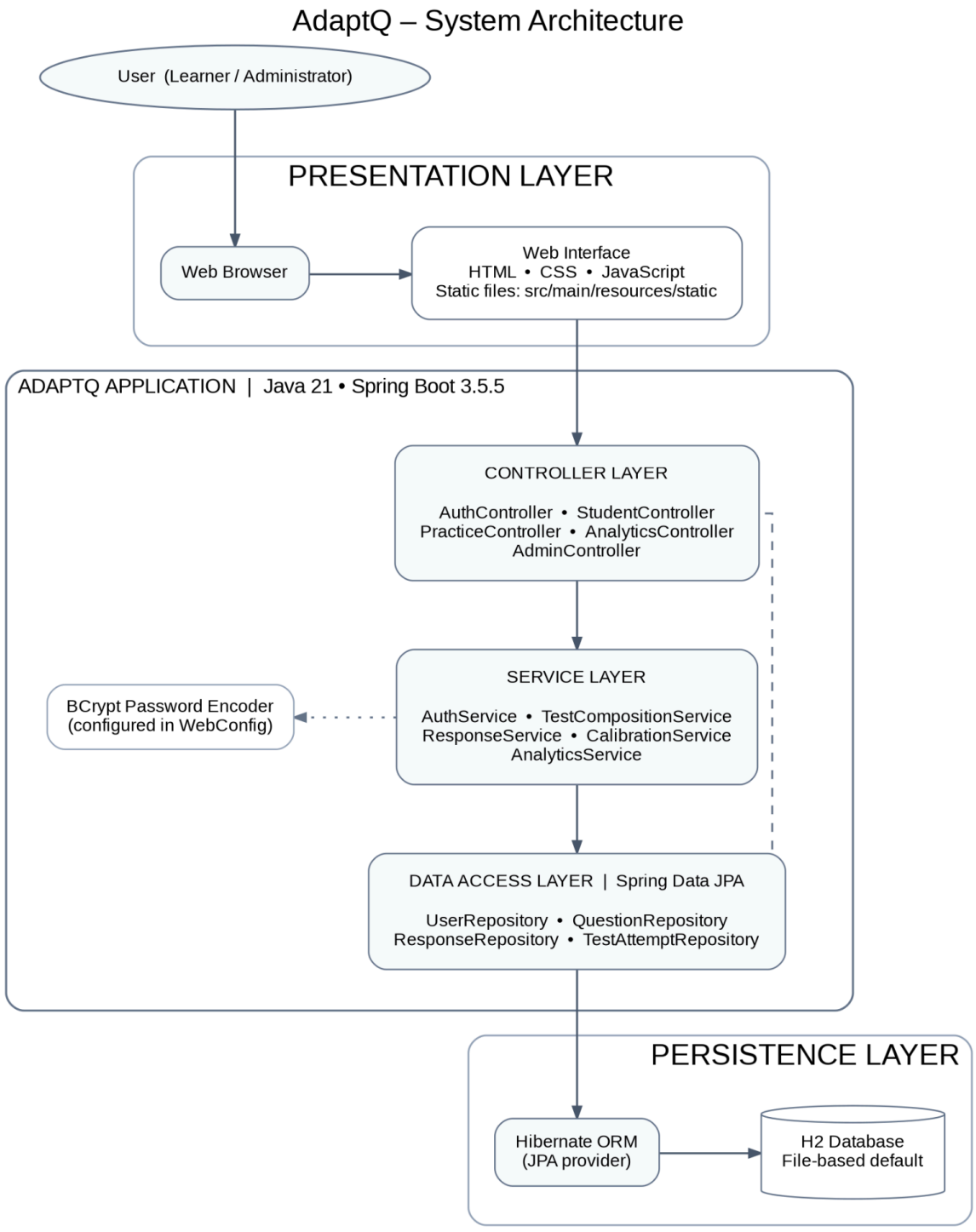


Figure 4.1: High-Level System Architecture of AdaptQ

The architecture diagram provides an overview of the principal application components and their relationships. It helps explain how the presentation, application, and persistence layers work together to support the functionality of AdaptQ.

4.8.2 Adaptive Quiz Activity Diagram

The activity diagram represents the adaptive quiz workflow from the initial request for a question through question selection, presentation, response submission, and updates to the relevant estimates.

The workflow begins when the learner starts or continues adaptive practice. The system identifies eligible questions and uses the learner's current ability estimate and question-difficulty information to guide selection.

After a question is presented, the learner submits an answer. The backend processes the response and coordinates the relevant updates. The workflow then determines whether the practice session should continue or end.

The diagram should also represent the situation in which no suitable question is available, allowing the workflow to terminate without attempting to select a question that cannot be provided.

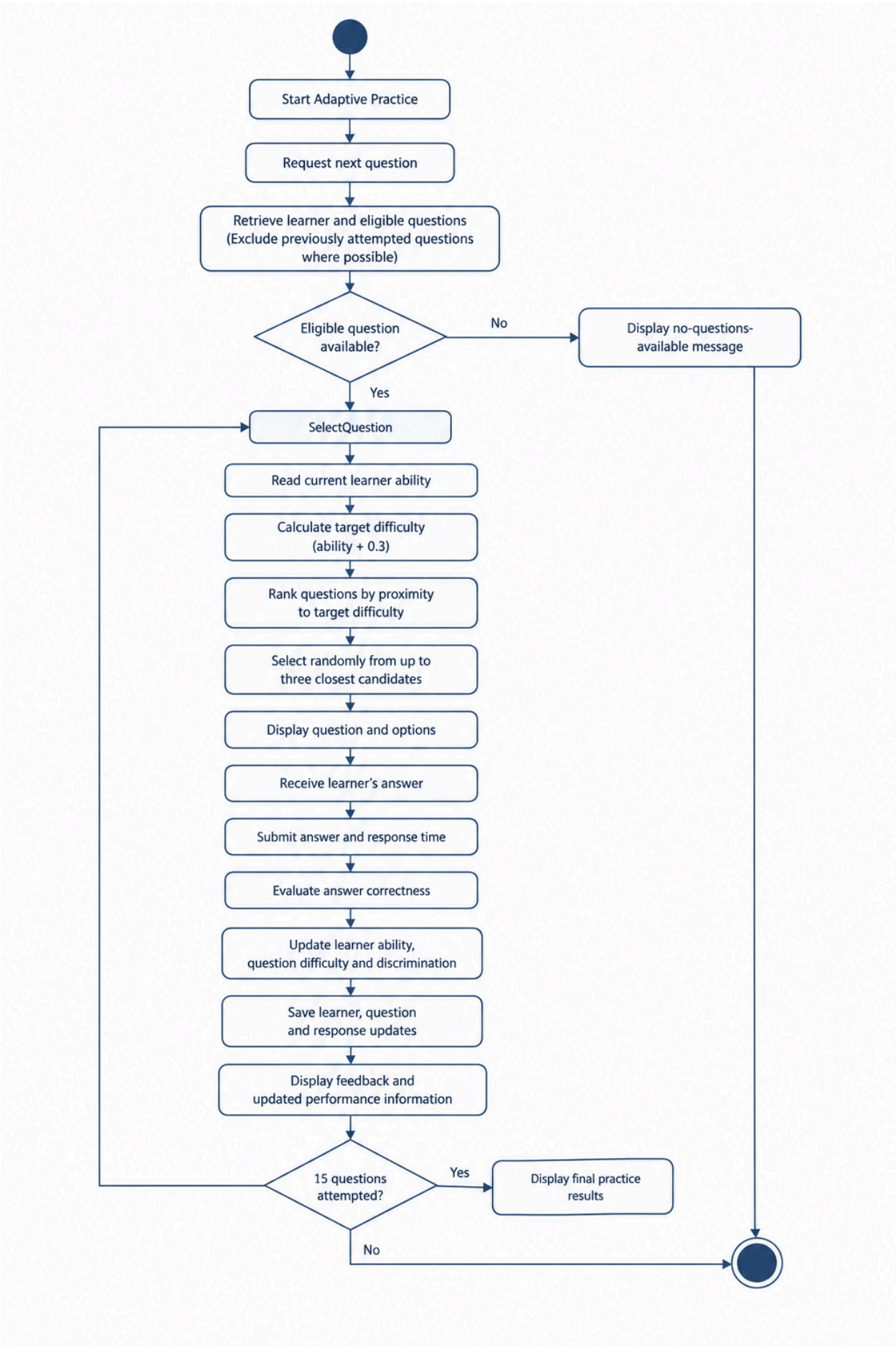

Start Adaptive Practice
Request next question
Retrieve learner and eligible questions (Exclude previously attempted questions where possible)
Eligible question available?
No
Display no-questions-available message
Yes
SelectQuestion
Read current learner ability
Calculate target difficulty (ability + 0.3)
Rank questions by proximity to target difficulty
Select randomly from up to three closest candidates
Display question and options
Receive learner's answer
Submit answer and response time
Evaluate answer correctness
Update learner ability, question difficulty and discrimination
Save learner, question and response updates
Display feedback and updated performance information
15 questions attempted?
Yes
Display final practice results
No

Figure 4.2: Activity Diagram of the Adaptive Quiz Workflow

The activity diagram provides a process-oriented view of the adaptive quiz mechanism. It illustrates the order of activities and the decisions that control the continuation or termination of practice.

4.8.3 UML Sequence Diagram of a Quiz Interaction

The sequence diagram illustrates the order of interactions involved in a quiz question.

The interaction begins when the learner uses the frontend to request a question. The backend controller handles the request and coordinates with the relevant service components to obtain a suitable question. The selected question is then returned to the frontend for presentation.

After the learner submits an answer, the backend processes the response and coordinates the relevant updates to learner and question information. The resulting information can be used to guide subsequent question selection.

The sequence diagram should show the learner, frontend, relevant backend controller, service components, and database as participants where supported by the actual implementation. The precise message sequence and responsibilities must match the application's source code.

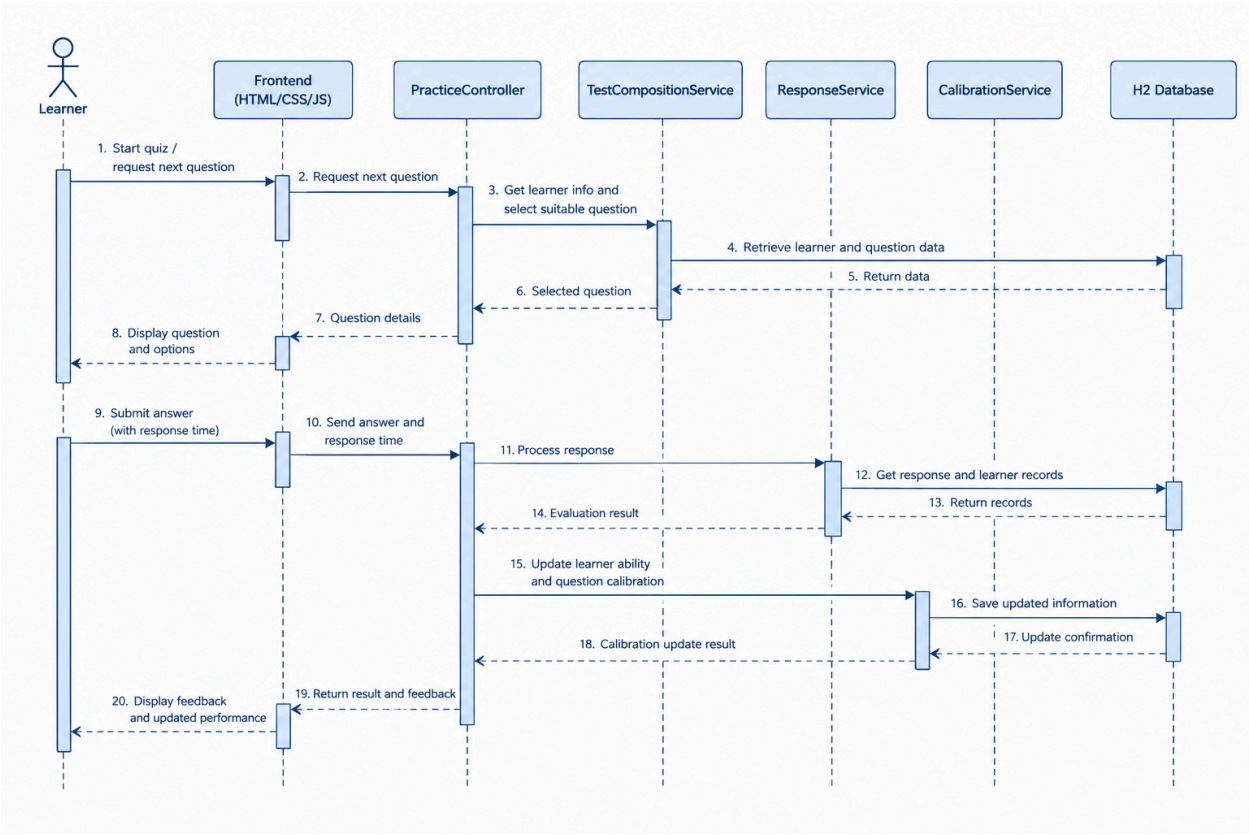


Figure 4.3: UML Sequence Diagram of a Quiz Interaction in AdaptQ

The sequence diagram provides a time-ordered view of the interaction between the frontend and backend components. It complements the architecture diagram by showing the order in which requests and responses occur during a quiz interaction.

4.9 Summary

This chapter described the system requirements, technology stack, architecture, adaptive quiz design, question bank, and principal implementation components of AdaptQ.

The application combines a browser-based interface with a Java and Spring Boot backend and a relational persistence layer. Its adaptive functionality uses learner-response information, estimated learner ability, and question-difficulty estimates to guide question selection.

The chapter also consolidated the main system design diagrams into a dedicated section. The architecture diagram illustrates the relationship between the major application layers, the activity diagram represents the adaptive quiz workflow, and the sequence diagram describes the interaction between the learner and the application components during quiz practice.

The modular organisation provides a foundation for maintaining the application's components and extending its capabilities. However, the effectiveness of adaptive selection depends on the

quality of the question bank, the behaviour of the calibration mechanism, and the availability of sufficient response data.

The next chapter presents the results and overall observations.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the results and observations from the development of AdaptQ, an adaptive quiz application designed to support learning through interactive assessments. It discusses the application's startup process, user authentication, quiz functionality, adaptive question selection, performance analytics, administrative dashboard, and additional features.

The chapter also examines the usefulness of the implemented features, the role of adaptive question selection in supporting personalised practice, and the limitations that may affect the application's effectiveness.

Screenshots of the running application are included to demonstrate its interface and principal functionalities.

5.2 Application Startup and Initial Interface

The application is developed as a web application using Java and Spring Boot for backend processing and HTML, CSS, and JavaScript for the frontend. The application can be launched in a suitable development environment and accessed through a web browser.

Successful startup makes the application's interface and backend functionality available for interaction. The browser-based approach also avoids requiring users to install a separate desktop application.

The initial interface provides an entry point through which users can access the available application features.

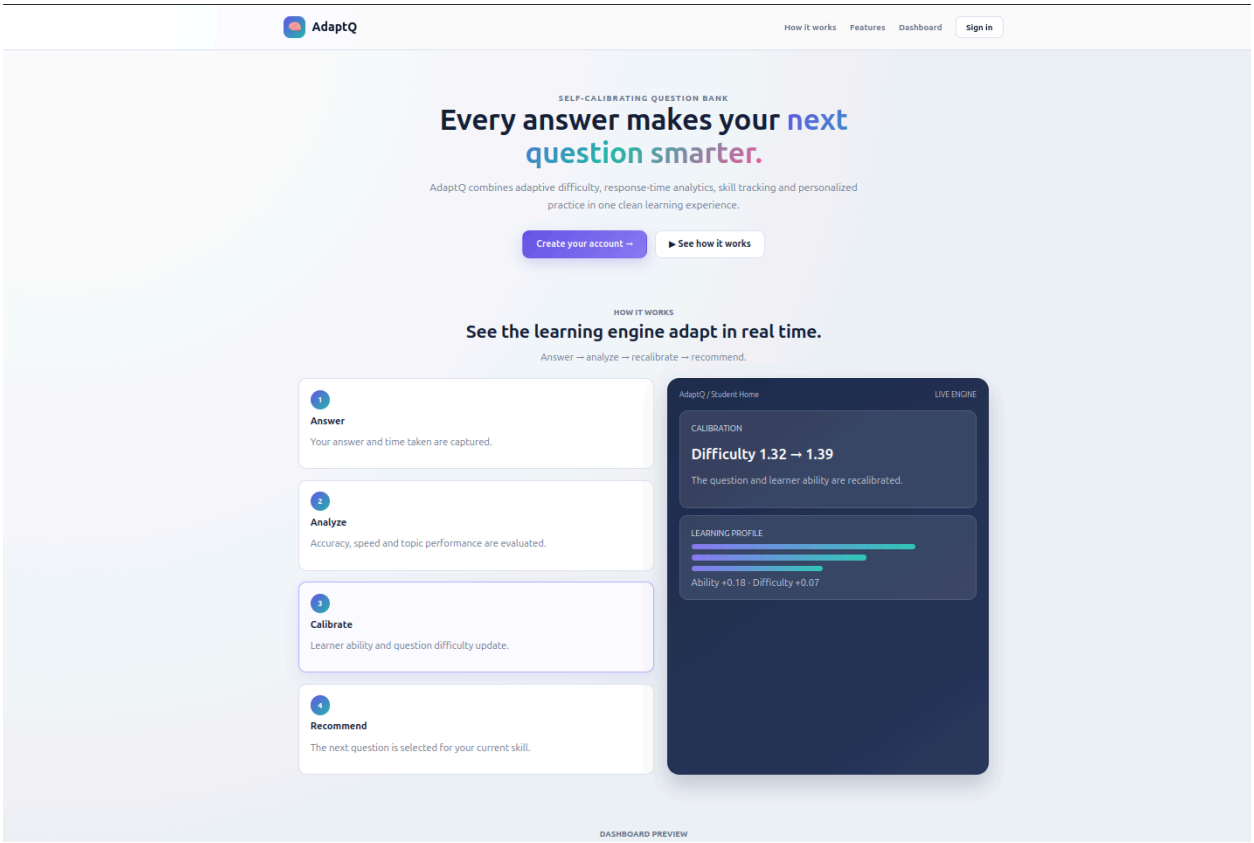


Figure 5.1: Initial Interface of the AdaptQ Application

The interface provides the starting point for interacting with the quiz system. Its web-based structure allows the application to present quiz-related information and user options through a browser.

5.3 User Authentication and Account Access

AdaptQ provides login and registration functionality to support user access. Registration allows a new user to create an account, while login provides access to the application using the relevant account credentials.

Authentication is important for applications that maintain learner-specific information because it provides a mechanism for associating quiz activity and performance records with individual users.

The availability of account access also supports the separation of learner activities from administrative operations where the corresponding access controls are implemented.

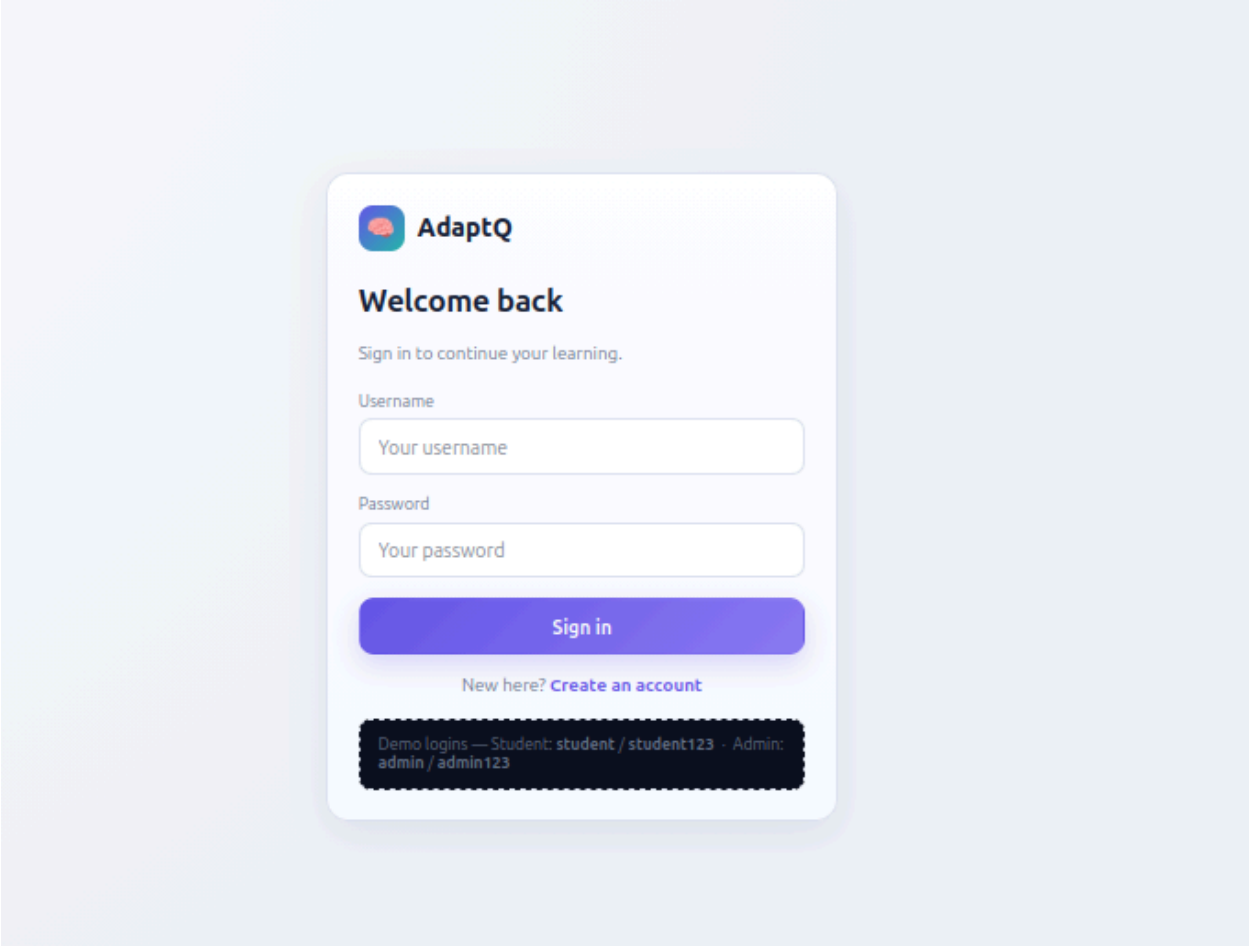


Figure 5.2: Login Interface

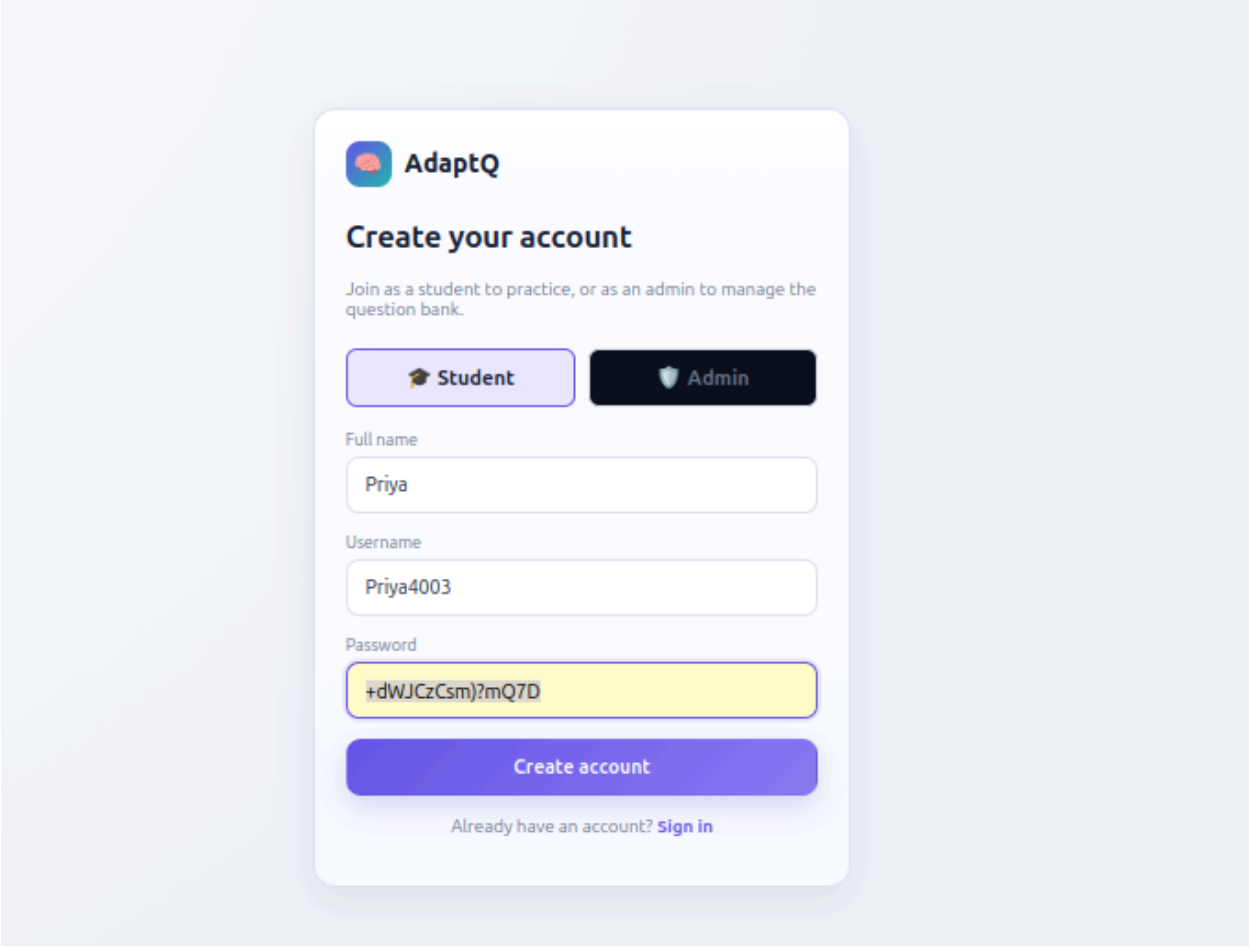


Figure 5.3: User Registration Interface

The login and registration interfaces form part of the application's access workflow. They provide users with a means of entering the system before interacting with its available functionality.

5.4 Quiz Functionality

Quiz functionality is one of the principal components of AdaptQ. Users can begin a quiz, view questions, select answers, and submit their responses for processing.

The backend processes submitted answers and determines whether they are correct. This information can then contribute to the calculation of quiz results and the updating of information used by the adaptive mechanism.

The availability of interactive quizzes allows learners to practise different computing subjects through a common interface.

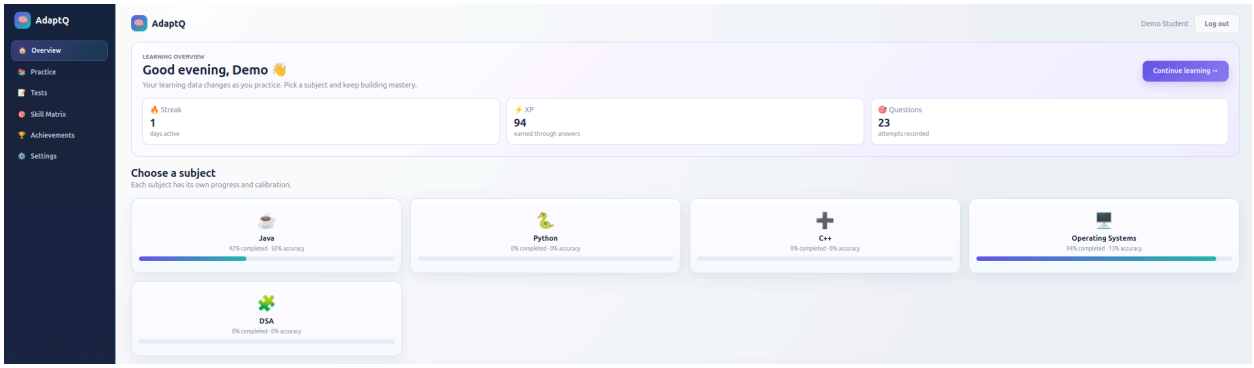


Figure 5.4: Quiz or Subject Selection Interface

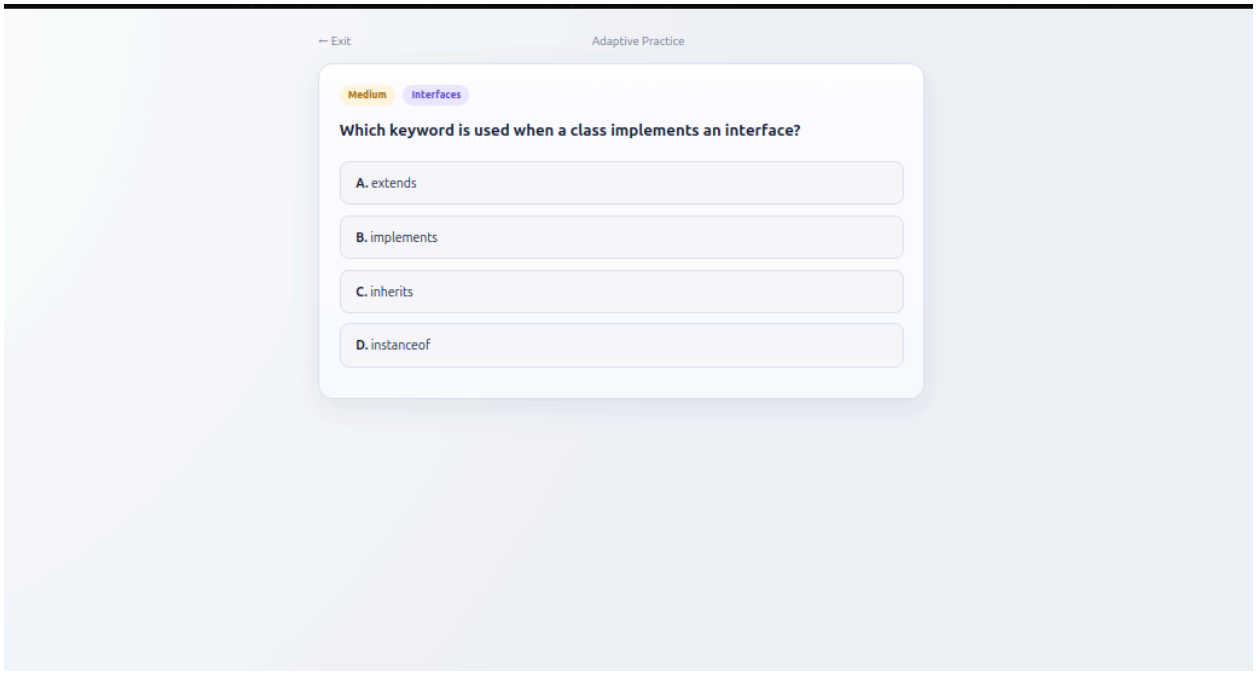


Figure 5.5: Quiz Question Interface

The question interface represents the main interaction between the learner and the application. By submitting answers, learners generate the response information required for scoring and adaptive question selection.

5.5 Adaptive Question Selection

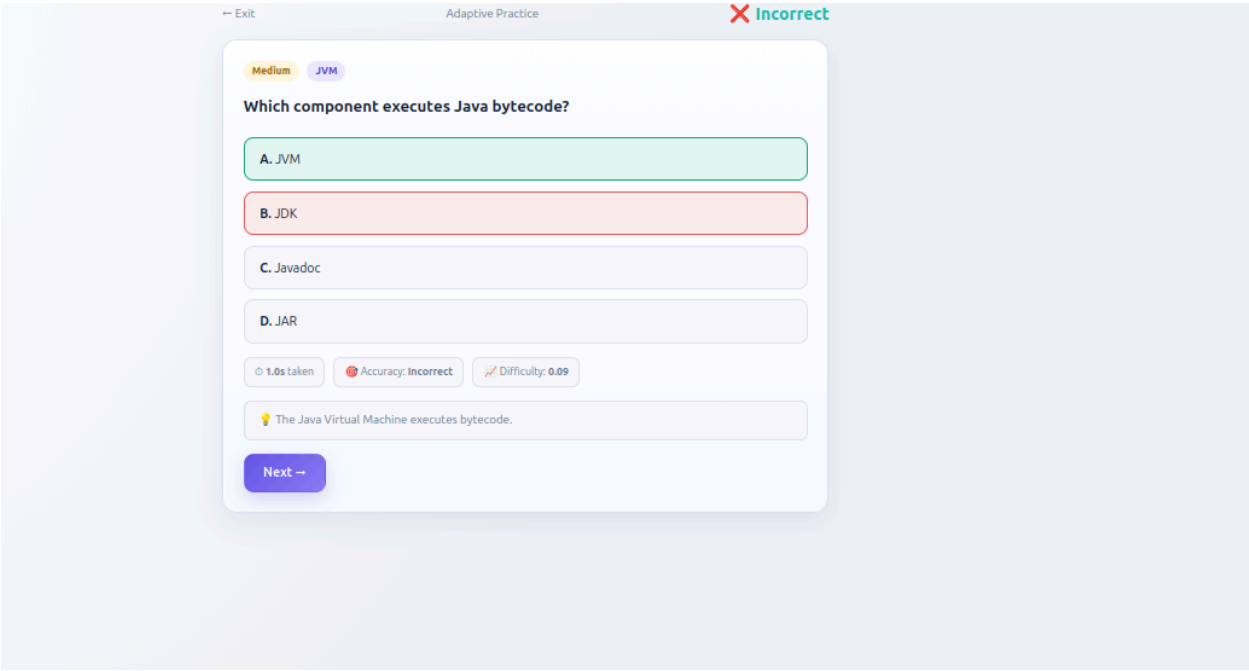
AdaptQ incorporates an adaptive mechanism intended to adjust question selection according to learner performance.

When a learner submits an answer, the application processes the response and uses relevant information to update its internal estimates. The adaptive mechanism considers estimated learner ability and question difficulty when selecting subsequent questions.

The selection process can be summarised as follows:

1. The system obtains the learner's current ability estimate.
2. It identifies eligible questions from the available question bank.
3. It calculates a target difficulty using the learner estimate and a selection adjustment.
4. It identifies questions whose estimated difficulty is close to the target.
5. It selects a suitable question while attempting to avoid repetition.
6. The learner submits an answer.
7. The system processes the response and updates relevant estimates.
8. The updated information guides subsequent question selection.

This approach differs from a quiz that follows an entirely fixed sequence of questions. Instead, the selection process uses information from previous responses to guide later questions.



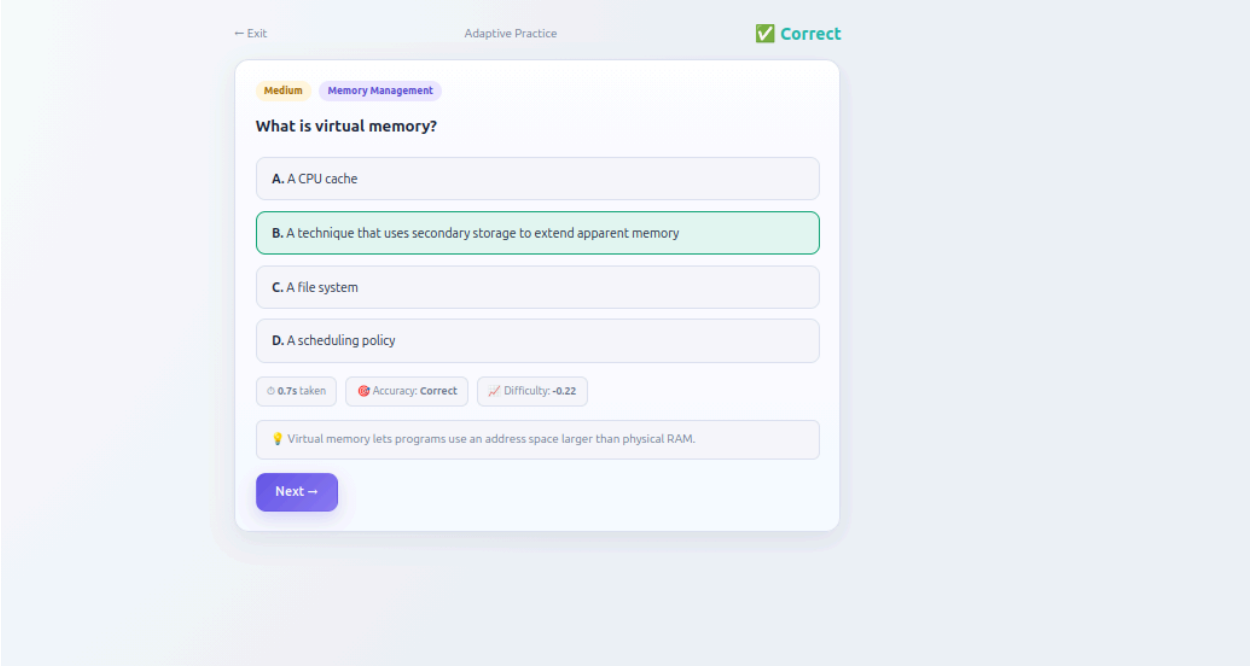


Figure 5.6: Example of Question Selection During Adaptive Practice

The adaptive mechanism should be understood as an implementation of performance-informed question selection. Its effectiveness depends on the available question bank, the accuracy of the internal estimates, and the behaviour of the calibration logic. The implementation has not been established as a professionally validated computerised adaptive testing system.

5.6 Results and Performance Analytics

AdaptQ provides functionality for displaying quiz results and performance-related information. Such information helps learners understand the outcomes of their practice and review their performance.

Performance information can include scores and other statistics made available by the application. These results provide a summary of quiz activity and can help learners identify areas that may require further practice.

Performance analytics also complement adaptive question selection. While the adaptive mechanism uses response information internally, the analytics interface presents relevant information to users in a more accessible form.

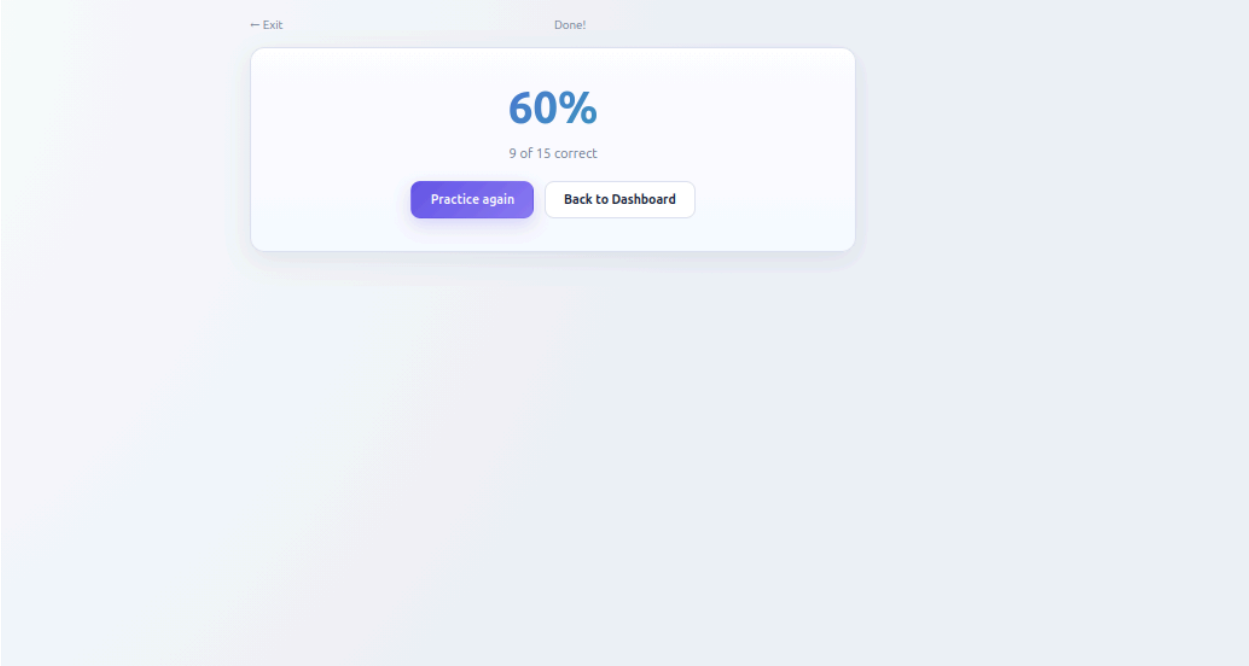


Figure 5.7: Quiz Results Interface

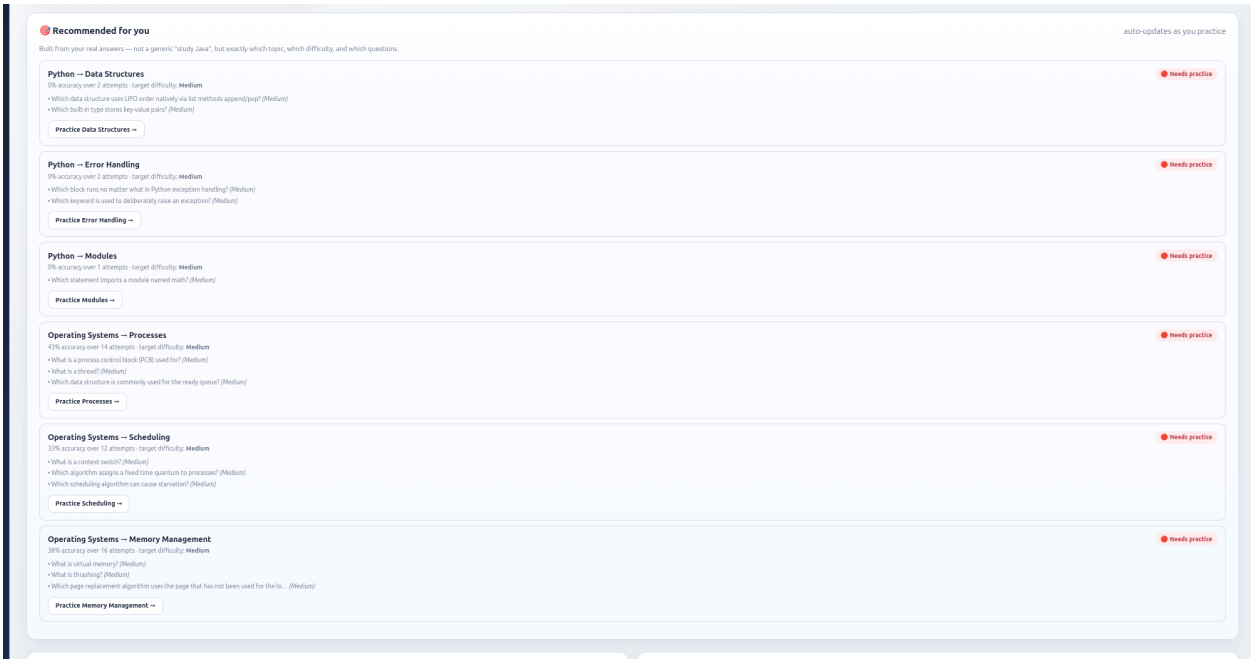


Figure 5.8.1: Further Practice Dashboard

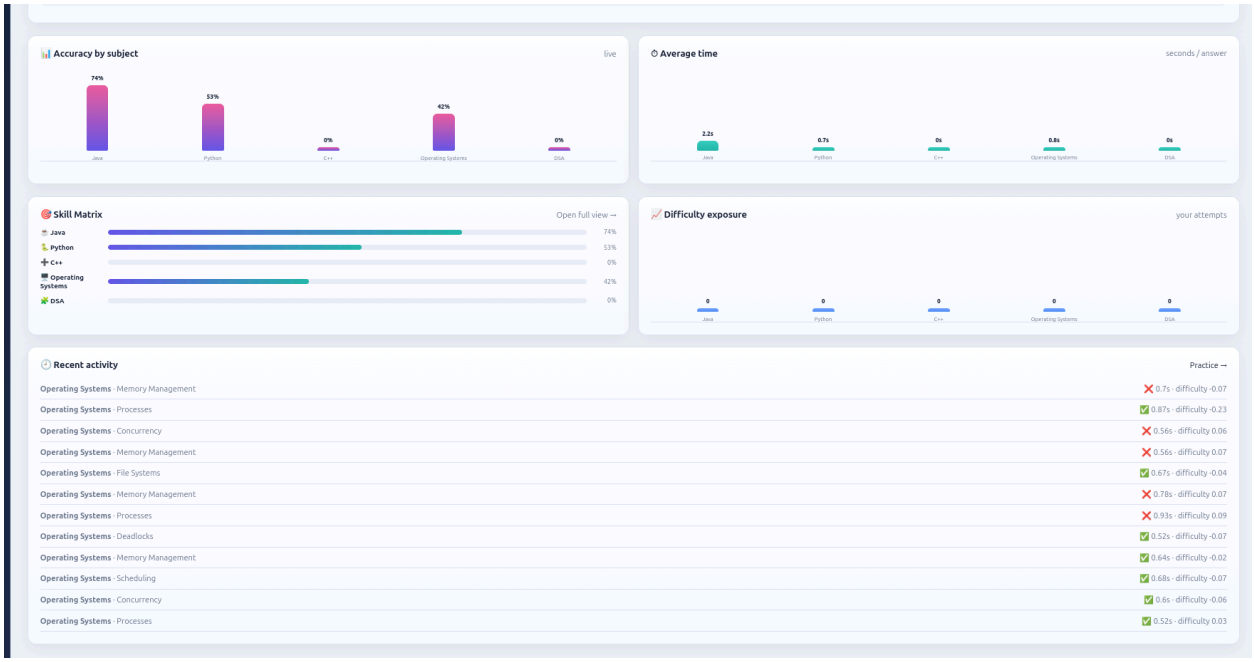


Figure 5.8.2: Performance Analytics Interface

The availability of these features provides users with feedback about their quiz activity. More detailed progress tracking over longer periods could further improve the usefulness of the application.

5.7 Administrative Dashboard

AdaptQ includes an administrative interface that provides access to information relating to the question bank and other aspects of the application.

Administrative functionality is relevant because an adaptive quiz system depends on the quality and organisation of its questions. Question-related information can help administrators review the available content and observe information associated with question difficulty and learner activity.

The administrative interface also provides a different perspective from the learner-facing quiz pages. While learners primarily interact with questions and results, administrative views focus on managing or examining information associated with the system.

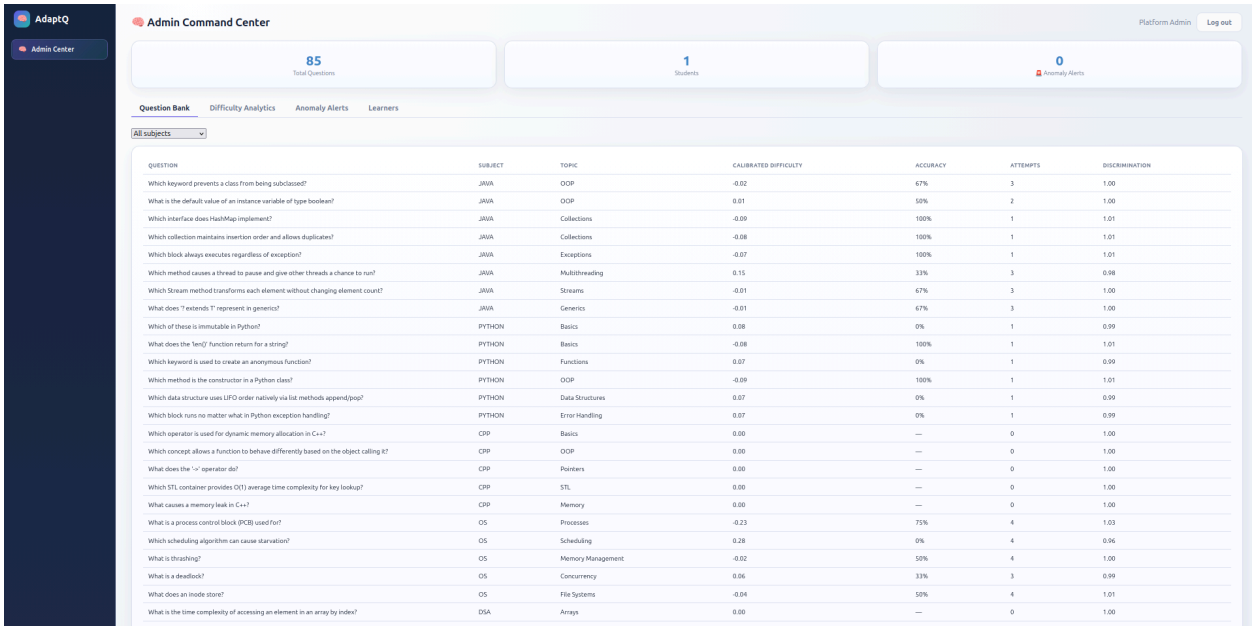


Figure 5.9: Administrative Dashboard

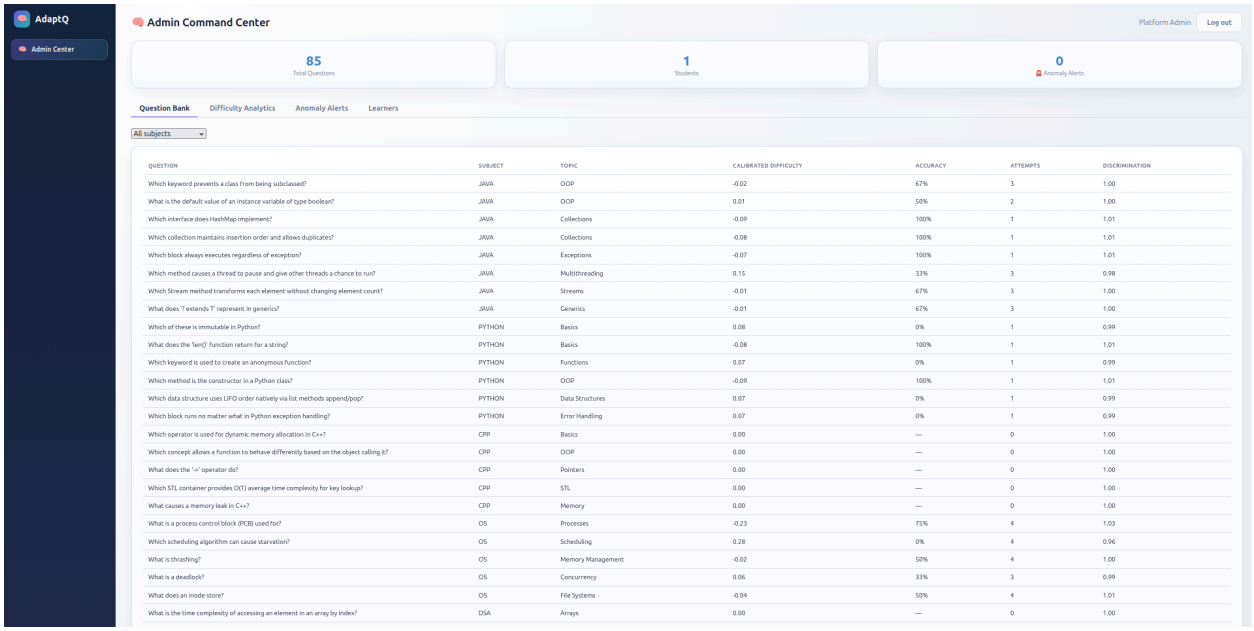


Figure 5.10.1 Question Bank

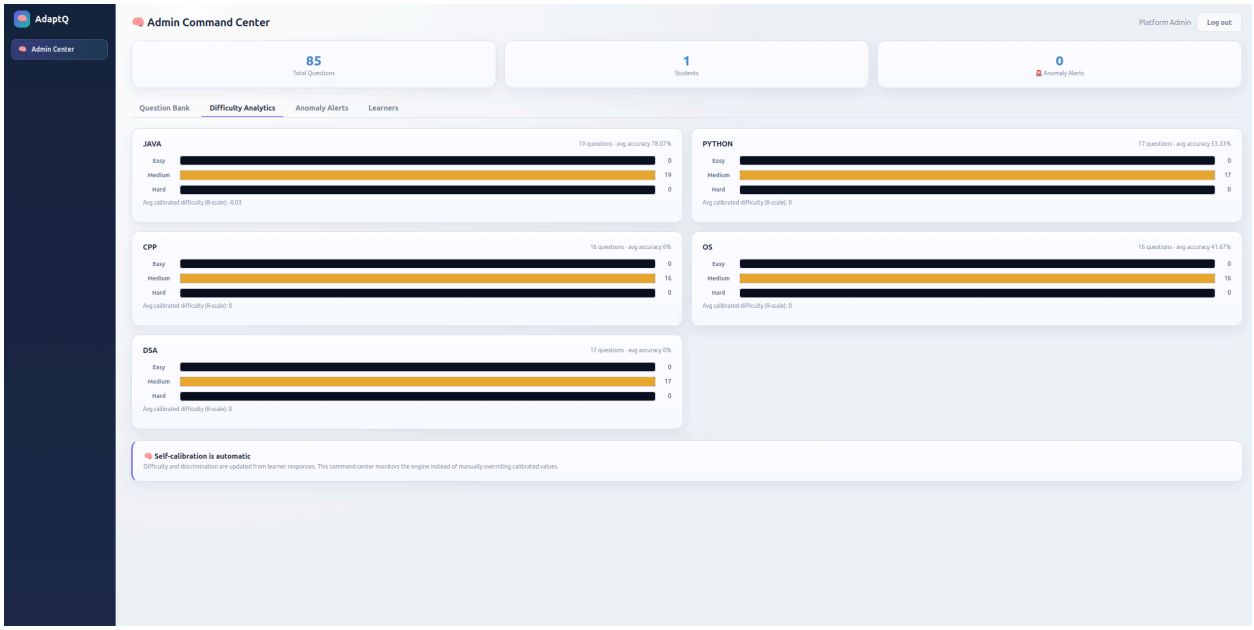


Figure 5.10.2 Difficulty Analytics

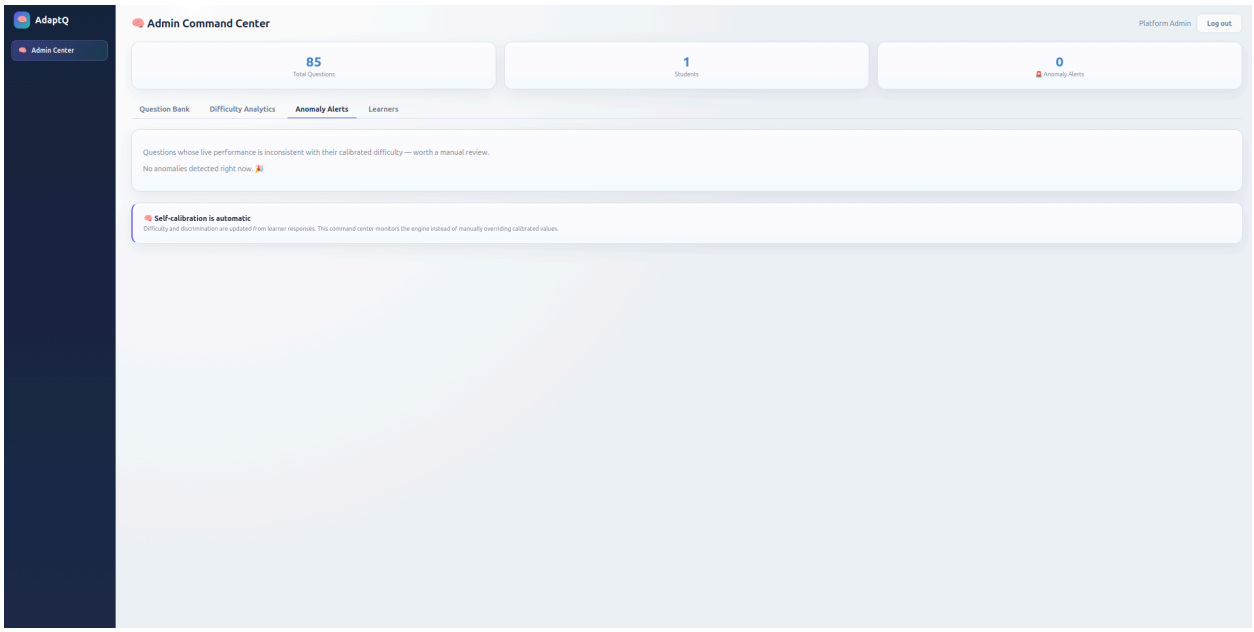


Figure 5.10.3 Anomaly Alerts

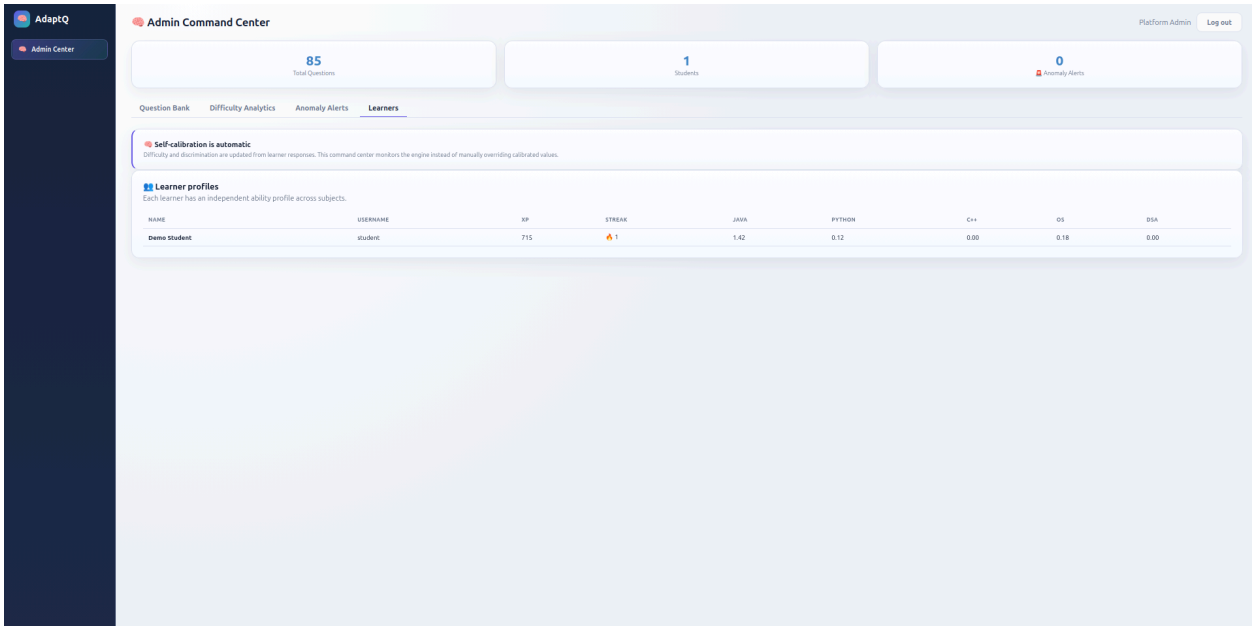


Figure 5.10.4 Learners

The usefulness of the administrative dashboard depends on the information and actions exposed through its interface. Additional question-management workflows and more comprehensive administrative controls could improve the system's maintainability.

5.8 Additional Features

In addition to its core quiz functionality, AdaptQ includes features intended to make practice more engaging and provide users with alternative ways to interact with the application.

5.8.1 XP and Achievements

XP and achievements introduce elements of gamification into the learning experience. Such features can provide users with visible indicators of activity or progress and make repeated practice more engaging.

Gamification is intended to complement the learning process rather than replace the educational value of the questions themselves.

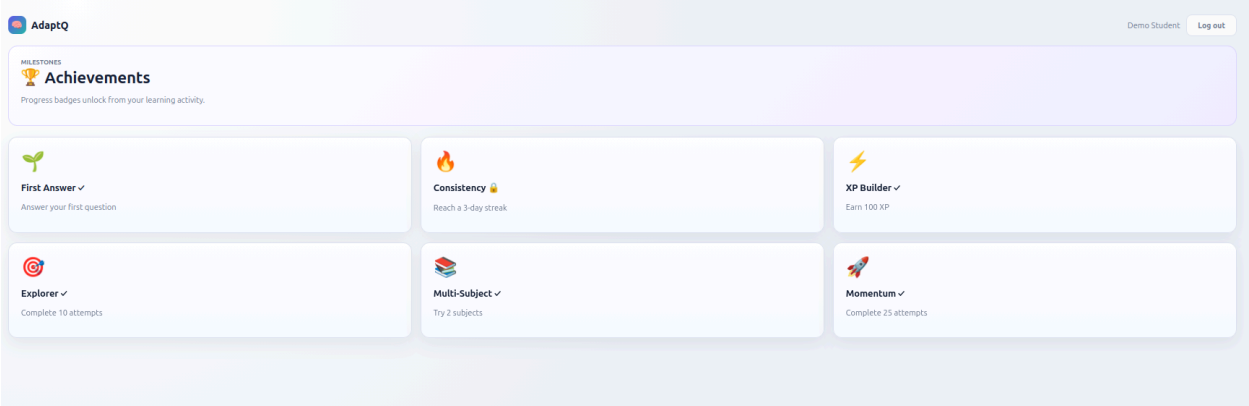


Figure 5.11: XP and Achievements Interface

5.8.2 Time Attack

Time Attack provides a timed challenge format for quiz practice. Unlike an untimed practice session, a timed activity introduces a time constraint that can encourage learners to practise answering questions within a limited period.

This feature offers an alternative to regular adaptive practice and may appeal to learners who prefer a more challenging assessment format.

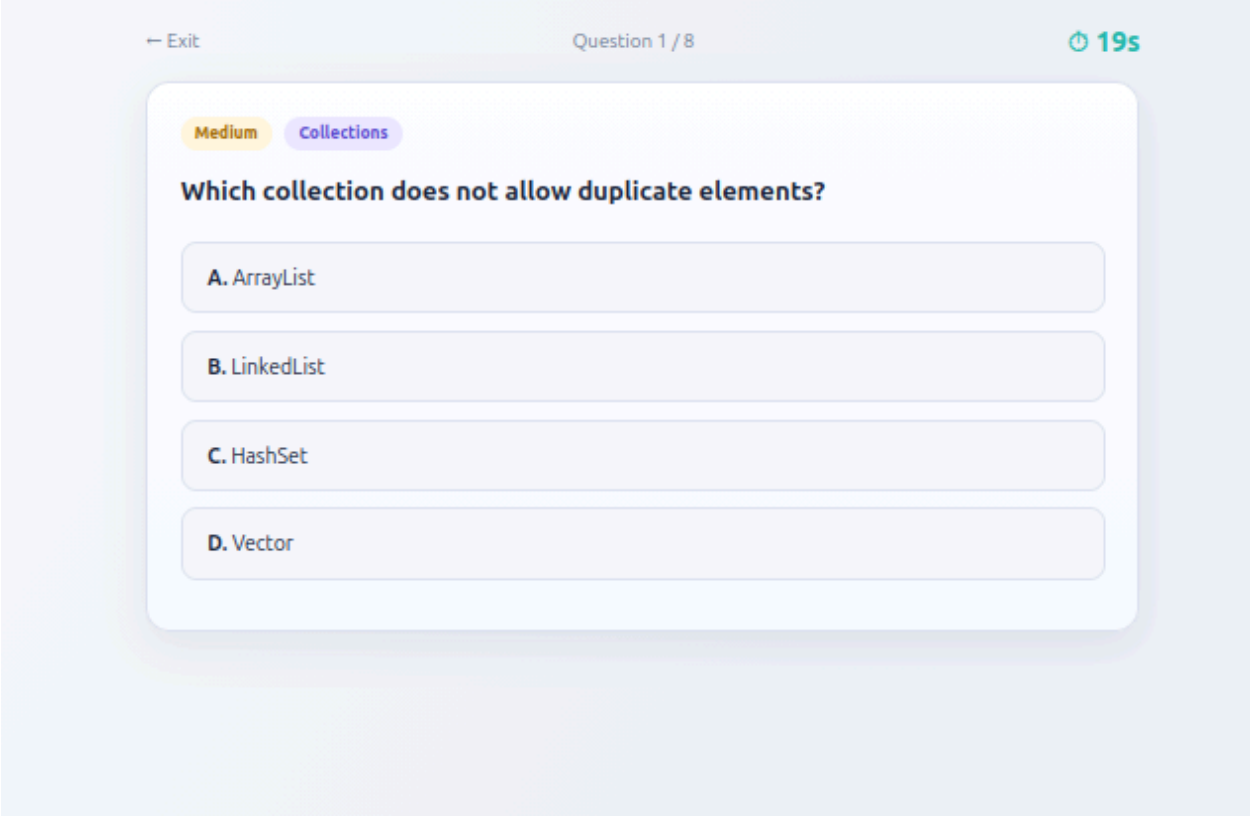


Figure 5.12: Time Attack Mode

5.8.3 Test Mode

Test Mode provides learners with an alternative way to interact with the quiz system by allowing them to practise or assess their knowledge through a test-oriented experience. Unlike features primarily intended to encourage engagement, this mode focuses on answering questions and reviewing performance within a structured assessment activity.

The inclusion of Test Mode broadens the ways learners can use AdaptQ, supporting practice alongside the application's adaptive quiz workflow. Its usefulness depends on the behaviour implemented in the application, including how questions are presented, how responses are evaluated, and what feedback is provided after the test.

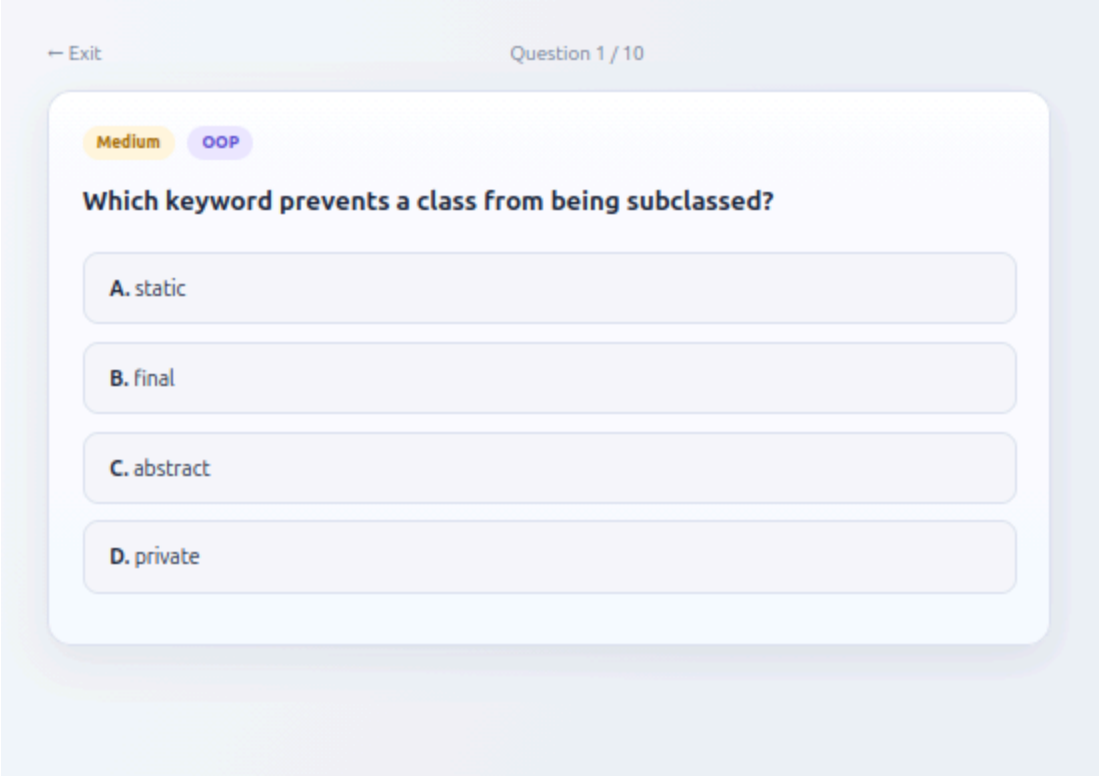


Figure 5.13: Test Mode

5.8.4 Skill Matrix

The Skill Matrix provides a structured view of a learner's performance across different skills or subject areas. It helps organise performance information so that learners can identify areas in which they perform well and areas that may require further practice.

When considered alongside quiz results and performance analytics, the Skill Matrix can help learners interpret their progress across the areas represented in the application. The value of this feature depends on the performance data available and how accurately that data reflects the learner's demonstrated knowledge.

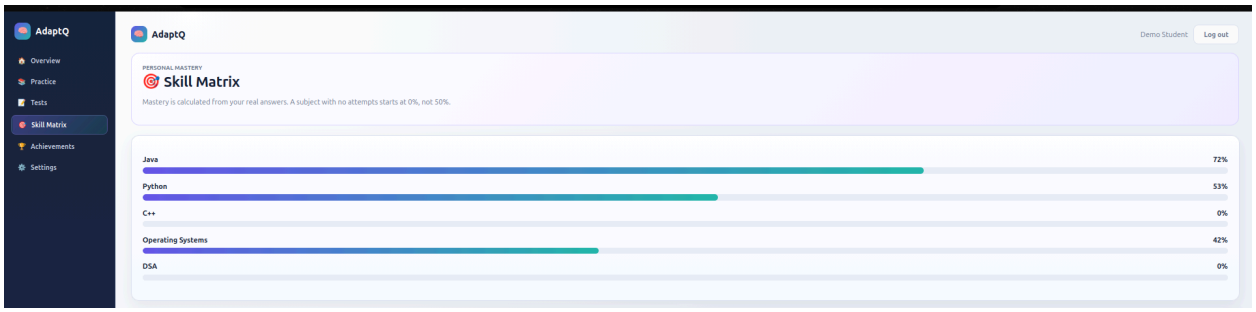


Figure 5.14: Skill Matrix

5.8.5 Integration with the Main Learning Experience

The additional features complement the application's main quiz workflow by providing alternative forms of interaction, assessment, and feedback.

Test Mode supports a structured testing experience, while the Skill Matrix presents performance across different skills or subject areas. XP and achievements focus on engagement, whereas Time Attack introduces a timed activity. Together with quiz results and performance analytics, these features contribute to a broader learning and practice experience.

The educational value of these features would benefit from further evaluation through sustained use, learner feedback, and assessment of their effectiveness in supporting practice and understanding.

5.9 Discussion

The development of AdaptQ brings together several components required for an interactive quiz application. These include a web-based interface, account access, quiz processing, adaptive question selection, performance information, administrative functionality, and additional engagement features.

One of the principal design objectives is to use learner performance to guide subsequent question selection. This provides a foundation for personalised quiz practice rather than relying exclusively on a predetermined sequence of questions.

The application also provides a means of presenting quiz results and performance-related information. This is useful because learners need feedback about their attempts in addition to the questions themselves.

The administrative interface supports the examination of question-bank information and related statistics. Since adaptive selection depends on question quality and difficulty estimates, maintaining suitable question content is an important consideration.

However, several limitations should be acknowledged. The quality of adaptive selection depends on the number and variety of available questions, the accuracy of the calibration mechanism, and the response information collected during practice. Limited question coverage may reduce the range of questions available for a particular learner or subject.

Similarly, the presence of performance analytics does not, by itself, establish that the application improves learning outcomes. Demonstrating such an improvement would require a more systematic evaluation involving learners, defined assessment criteria, and sufficient data.

The additional features provide further opportunities for engagement, but their effectiveness would also benefit from evaluation. Future work could examine usability, learner satisfaction, question quality, and the consistency of adaptive selection.

5.10 Chapter Summary

This chapter discussed the principal functionality of AdaptQ, including application startup, account access, quiz interaction, adaptive question selection, performance analytics, administrative functionality, XP and achievements, and Time Attack.

The application combines these features within a browser-based interface supported by a Java and Spring Boot backend. Its adaptive mechanism uses learner-response information and question-related estimates to guide subsequent question selection.

Although the implemented features provide a foundation for adaptive quiz practice, further development and evaluation would be necessary to assess learning effectiveness, improve calibration, expand question coverage, and strengthen progress tracking.

The next chapter concludes the report and discusses potential directions for future development.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Introduction

This chapter presents the conclusion of the AdaptQ project, summarises its main contributions, discusses the limitations of the current implementation, and identifies possible areas for future development.

AdaptQ was developed as an adaptive quiz application intended to make practice more responsive to individual learner performance. The system combines subject-based quizzes, performance tracking, adaptive question selection, and gamification features within a web-based application. The project also incorporates administrative functionality for managing and examining aspects of the question bank.

6.2 Project Conclusion

The primary objective of AdaptQ was to develop a quiz application that offers more than a fixed sequence of questions. Conventional quiz systems often present questions according to a predetermined sequence or allow learners to select questions based on predefined categories. AdaptQ explores an alternative approach by using information about learner responses to influence subsequent question selection.

The application incorporates an adaptive mechanism that considers learner performance and question difficulty. Response information, including correctness and response time where applicable, contributes to the system's assessment of performance. This information is used to guide question selection during practice.

The project also brings together several supporting features, including user authentication, subject-based practice, result reporting, performance analytics, and gamification elements such as experience points, achievements, and Time Attack activities. These features provide different ways for learners to interact with the application and review their progress.

From an implementation perspective, the project demonstrates the integration of a web-based interface with application logic and persistent data storage. The use of Java, Spring Boot, Maven, and a database-backed persistence layer provides the foundation for the application's principal operations.

Overall, AdaptQ represents an implementation of an adaptive quiz application that combines assessment, performance monitoring, and engagement-oriented features in a single system. The completed application provides a foundation for further refinement of its adaptive logic, question management processes, and learning analytics.

6.3 Achievement of Project Objectives

The project addressed several objectives during development.

6.3.1 Development of a Web-Based Quiz Application

A web-based application was developed to provide access to quiz functionality through a user interface. The application includes user access functionality, subject-based practice, question presentation, and result handling.

6.3.2 Implementation of Adaptive Question Selection

The system incorporates an adaptive question-selection mechanism that uses learner performance and question difficulty information to guide the selection of subsequent questions. This provides an alternative to presenting every learner with an identical question sequence.

6.3.3 Performance Tracking and Analytics

The application includes result and performance-related features intended to help learners review their quiz activity. Administrative features also provide ways to examine question-bank information and related analytics.

6.3.4 Integration of Gamification Features

Features such as experience points, achievements, and Time Attack activities were incorporated to provide additional forms of interaction beyond standard quiz attempts.

6.3.5 Administrative Functionality

Administrative functionality supports aspects of question-bank management and the examination of question-related information. These capabilities provide a basis for maintaining the content used by the quiz application.

Taken together, these objectives establish the principal functional areas of the AdaptQ application. However, implementing these features does not by itself establish that the adaptive mechanism improves learning outcomes. That question requires further evaluation.

6.4 Limitations of the Current System

Although AdaptQ provides a working foundation for adaptive quiz practice, several limitations remain.

6.4.1 Limitations of the Adaptive Algorithm

The adaptive mechanism uses performance and question-difficulty information to guide question selection. Its effectiveness depends on the quality of the available data and the accuracy of the underlying estimates.

The current implementation should not be considered equivalent to a formally validated computerized adaptive testing system. Further work would be required to establish the reliability, accuracy, and educational suitability of its adaptation process.

6.4.2 Dependence on Question-Bank Quality

The usefulness of an adaptive quiz system depends heavily on the quality of its questions. Questions must be accurate, relevant to the selected subject, and assigned appropriate difficulty levels.

Incorrect answers, inconsistent difficulty classifications, ambiguous wording, or insufficient topic coverage may reduce the usefulness of the system. Question-bank quality therefore remains an important consideration for future development.

6.4.3 Limited Evidence of Educational Effectiveness

The availability of adaptive quizzes and performance analytics does not automatically demonstrate an improvement in learning. Establishing educational effectiveness would require structured testing involving learners, clearly defined evaluation criteria, and analysis of the collected results.

The current project focuses primarily on developing and demonstrating the application. Its effectiveness as a learning intervention has not been established through a controlled educational study.

6.4.4 Scope of Performance Analytics

The application provides performance-related information, but the depth and usefulness of such analytics depend on the data collected and the analysis performed.

More advanced reporting could provide clearer information about topic-level weaknesses, changes in performance over time, and areas requiring additional practice.

6.4.5 Scope for Interface and Usability Improvements

As with many software projects developed within academic time constraints, the interface and user experience can be refined further. Additional usability testing would help identify confusing interactions, improve navigation, and ensure that the application remains accessible to users with different levels of technical familiarity.

6.4.6 Deployment and Scalability Considerations

The application's suitability for a larger number of concurrent users would need to be evaluated through additional deployment and performance testing. Database configuration, server resources, security practices, and application monitoring would also require attention before the system could be treated as a production-ready service.

6.5 Future Scope

Several improvements could extend the functionality and reliability of AdaptQ.

6.5.1 Improvement of the Adaptive Algorithm

The adaptive question-selection mechanism could be refined to use more systematic methods for estimating learner ability and question difficulty. Future versions could investigate established approaches to computerized adaptive testing, provided the necessary assumptions and data requirements are satisfied.

The selection process could also be evaluated to determine whether the questions presented are appropriately challenging and whether the adaptation responds consistently to changes in learner performance.

6.5.2 Expansion and Validation of the Question Bank

The question bank could be expanded to include more subjects, topics, and difficulty levels. Each question could be reviewed for accuracy, clarity, relevance, and appropriate difficulty classification.

A more structured question-review process could also help identify poorly performing or ambiguous questions. This would improve the reliability of the content on which the adaptive mechanism depends.

6.5.3 Advanced Learning Analytics

Future versions could introduce more detailed reports showing performance across individual topics, progress over multiple attempts, and recurring areas of difficulty.

Visual summaries could help learners understand their progress, while administrators could use aggregated information to identify questions that may require review. Any such analytics should be based on clearly defined metrics and interpreted carefully.

6.5.4 Personalised Learning Recommendations

AdaptQ could be extended to recommend topics or practice activities based on a learner's previous performance. For example, a learner who repeatedly struggles with a particular topic could be encouraged to review that topic before attempting more difficult questions.

Such recommendations would need to be designed carefully so that learners receive useful guidance without being restricted unnecessarily to a narrow set of questions.

6.5.5 Enhanced Gamification

The existing gamification features could be extended through additional achievements, progress milestones, and optional challenges. These features could provide learners with further ways to track their activity and maintain interest in regular practice.

However, gamification should support the learning process rather than encourage learners to focus exclusively on points, speed, or rankings.

6.5.6 Improved Administrative Tools

Administrative functionality could be extended with more comprehensive question-editing workflows, bulk question import and export, improved question validation, and more detailed reports.

These improvements could make it easier to maintain a larger question bank and ensure that the content remains consistent and suitable for adaptive practice.

6.5.7 Security and Deployment Improvements

Future development could strengthen authentication, access control, input validation, error handling, and the protection of stored user information.

The application could also be prepared for deployment in a more demanding environment through appropriate database configuration, automated testing, logging, monitoring, backup procedures, and performance evaluation.

6.5.8 Evaluation with Learners

A structured evaluation involving actual learners would be an important next step. Such an evaluation could examine usability, perceived usefulness, question difficulty, and the consistency of the adaptive question-selection process.

A separate educational study could investigate whether adaptive practice produces better learning outcomes than a fixed question sequence. This would require a suitable study design, defined outcome measures, and analysis of the collected data. Conclusions about learning effectiveness should only be drawn from the resulting evidence.

6.6 Final Remarks

AdaptQ demonstrates the development of a web-based quiz application that combines adaptive question selection, performance tracking, administrative functionality, and gamification features. The project brings these components together to explore how quiz practice can respond to learner performance rather than relying entirely on a fixed sequence of questions.

The development process also illustrates the importance of coordinating interface design, application logic, data persistence, and question-bank management when building an integrated software system.

Although the current implementation has limitations in adaptive evaluation, question-bank quality assurance, analytics, and deployment readiness, these limitations provide clear directions for further work. Improvements to the adaptive mechanism, more comprehensive testing, and evaluation with learners could help establish the system's reliability and educational value.

In conclusion, AdaptQ provides a foundation for continued development of an adaptive quiz application. Its longer-term value will depend not only on the addition of new features but also on the quality of its questions, the reliability of its adaptive behaviour, and evidence demonstrating its usefulness to learners.

6.7 Chapter Summary

This chapter summarised the objectives and principal contributions of the AdaptQ project. It discussed the implementation of the web-based quiz application, its adaptive question-selection mechanism, performance tracking, gamification features, and administrative functionality.

The chapter also identified limitations relating to adaptive algorithm validation, question-bank quality, educational evaluation, analytics, usability, and deployment. Finally, it outlined future development opportunities, including improvements to adaptation, personalised recommendations, administrative tools, security, and structured learner evaluation.

These areas provide a basis for further development and assessment of AdaptQ beyond the scope of the current project.

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